Singular learning theory without coordinates

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Abstract

We rework the derivation of the asymptotic expansion of the Bayesian free energy in singular learning theory, following Watanabe [1], so that the statements and the limiting objects are free of local coordinates on the resolved parameter space. The key observation is that the leading coefficient of the asymptotic expansion of the partition function is a positive measure μ on the resolution manifold, definable directly as a Laurent coefficient of the zeta function of the model. Watanabe's singular integral estimates (Theorems 4.6–4.9 of [1]) are then subsumed, at the level of statements, into a single kernel convergence theorem, valid for a general class of insertions, from which the limit law of the normalized evidence and the free energy formula

$$F_n \;=\; n\beta S_n + \lambda \log n - (m-1)\log\log n + F_n^R, \qquad F_n^R \to -\log \mathcal{Y}(\xi) \text{ in law},$$

follow by soft arguments. Local charts survive only inside proofs; no partition-of-unity consistency conditions, essential families of coordinates, or chart constants appear in any statement. Convergence in law is treated under the index $s = 2$ moment conditions of [1], and convergence of expectations under the index $s = 4$ conditions; for the latter, the only ingredient beyond the uniform integrability machinery of [1] is a matching lower bound on the evidence, which the kernel theorem supplies through a deterministic insertion. The four error formulas for the Bayes and Gibbs generalization and training errors, and the equations of states, are then derived in the same framework: the Gibbs pair follows from the free energy asymptotics alone, by differentiating tilted free energies in their couplings, while the Bayes pair uses a posterior form of the kernel theorem together with a Gaussian integration by parts for the limit functionals, proved here on the support of the leading measure; the random-variable equations of states and the empirical-variance relations behind WAIC are obtained in the same framework. Beyond a declared background package — the resolution theorem, the standard form of the log likelihood ratio, and the empirical process theory of [1] — the derivation is self-contained; in particular the tilted moment bounds and cubic Taylor remainders behind the Bayes errors, obtained in [1] by chart-level integration by parts, are reproved by a soft argument combining nonnegativity of the renormalized energy with exponential insertions of optimized vanishing rate.

1 Introduction

Let $q(x)$ be a true distribution, $p(x \mid w)$, $w \in W \subset \mathbb{R}^d$, a statistical model, $\varphi(w)$ a prior, and let $D_n = \{X_1, \ldots, X_n\}$ be independent samples from q. The central object of singular learning theory is the *stochastic complexity* (minus log evidence) at inverse temperature $\beta > 0$,

$$F_n \;=\; -\log \int_W \prod_{i=1}^{n} p(X_i \mid w)^{\beta}\, \varphi(w)\, dw.$$

When the set of true parameters $W_0 = \{w : p(\cdot \mid w) = q\}$ is a point at which the Fisher information is nondegenerate, the Laplace approximation yields $F_n = n\beta S_n + \frac{d}{2}\log n + O_P(1)$. In singular models — mixtures, neural networks, reduced-rank regression, hidden Markov models — W_0 is a real analytic set with singularities, the Laplace approximation fails, and Watanabe's theory [1] shows that instead

$$F_n \;=\; n\beta S_n + \lambda \log n - (m-1)\log\log n + O_P(1), \tag{1}$$

where $(-\lambda)$ is the largest pole of the zeta function $\zeta(z) = \int K(w)^z \varphi(w)\,dw$ of the model, m is its order, and S_n is the empirical entropy of q. When the prior is positive near W_0, λ is the real log canonical threshold of the Kullback–Leibler function K and m its multiplicity.

The proof in [1] proceeds by resolution of singularities: a proper real analytic map $g : M \to W$ is chosen so that $K \circ g$ and the pulled-back prior are simultaneously normal crossing in each chart of a manifold M, and the partition function is then analyzed chart by chart. The chart-level analysis (Theorems 4.6–4.9 of [1]) is sharp but heavily coordinate-dependent: the limiting random variable is exhibited as a sum over an "essential family of local coordinates" of integrals involving chart constants γ_b, monomial weights y^μ, and a partition of unity, and the consistency of these data across charts must be arranged by hand (Theorem 6.5 of [1]).

The purpose of this note is to show that both the statements and the limit objects can be made coordinate-free, in the following precise sense.

(i) The leading coefficient of the asymptotic expansion of the partition function is a finite *positive measure* μ on M, definable in one line as a Laurent-type limit of the zeta function (Definition 3.1); its support Σ is the "essential locus" of the resolution. All chart constants and consistency conditions are absorbed into μ.

(ii) Watanabe's singular integral estimates are replaced by a single *kernel theorem* (Theorem 5.2): the renormalized functionals

$$\Lambda_n[F] \;=\; \frac{(m-1)!\,n^\lambda}{(\log n)^{m-1}} \int_M F\big(nK_g(u), u\big)\,\omega(du)$$

converge, with a quantitative rate, to $\int_0^\infty\!\int_\Sigma F(t,u)\,t^{\lambda-1}\,d\mu(u)\,dt$, for arbitrary insertions $F(t,u)$ that are C^1 in u with exponential decay in t. Watanabe's insertions K^p, and the insertions needed later in the theory (posterior moments, fluctuation functionals), are special cases; "general p" costs nothing.

(iii) The limit law of the evidence and the free energy formula (1) follow from (i)–(ii) by the continuous mapping theorem: the limiting random variable is

$$\mathcal{Y}(\xi) \;=\; \frac{1}{(m-1)!}\int_0^\infty\!\int_\Sigma t^{\lambda-1}\,e^{-\beta t+\beta\sqrt{t}\,\xi(u)}\,d\mu(u)\,dt,$$

a functional of the pair $(\mu, \xi|_\Sigma)$ alone, where ξ is the limiting Gaussian process of the standard form of the log likelihood ratio. The pair depends on the chosen resolution; the law of $\mathcal{Y}(\xi)$ does not (Remark 9.1).

Charts are not eliminated: they persist inside the proofs, where the exact evaluation of monomial state densities is performed, and for these local calculations we cite [1] rather than repeat them. What disappears is every chart-dependent statement, together with the bookkeeping needed to glue chart-level results.

Sections 3–6 treat convergence in law, which requires only the index $s = 2$ moment conditions of [1]; Section 7 derives the corresponding statements in expectation under the index $s = 4$ conditions. There the upper-bound-only character of the toolkit must be supplemented by a matching lower bound on the evidence; the point of the coordinate-free framework is that this bound is not a new chart computation (the role played by [1, Thms. 4.8, 4.10]) but an application of the kernel theorem to a deterministic insertion, combined with the positivity $\mu(M) > 0$. Section 8 proves the four error formulas for the Bayes and Gibbs generalization and training errors, and the equations of states, in mean and at the level of random variables, together with the limit law of the quartet and the empirical-variance relations underlying the widely applicable information criteria. A byproduct of the framework is that the Gibbs pair requires no posterior asymptotics at all: since (λ, m) do not depend on the inverse temperature, the $\lambda \log n$ terms cancel in temperature-differences of free energies, and the Gibbs errors follow by differentiating tilted free energies in their couplings, with convexity justifying the interchange of limits. For the Bayes pair, the localization between the full and the inner posterior — for which [1] deploys the tail machinery of [1, Lems. 6.1–6.3] — is proved here from the envelope lemma alone (Lemma 8.27), and the σ-tilted moment bounds with their cubic Taylor remainders — in [1], the chart-level integration by parts of [1, Lem. 6.5] — are proved internally (§8.6, Theorem 8.18 through Lemma 8.21). What is taken from [1] is thereby confined to the background package of §2; Remark 9.3 gives the inventory. Remark 9.4 delineates the boundary of the method.

The point of view that the coefficients of the asymptotic expansion of a Laplace-type integral are distributions supported on the critical locus, obtained as Laurent coefficients of $\int f^z \varphi$, with orders controlled by the drop in pole order, is classical in singularity theory; see Arnold–Gusein-Zade–Varchenko [3], Malgrange [5], and, for the complex-analytic counterpart, Barlet [6]. Section 9 contains precise pointers.

2 Background and standing assumptions

This section collects, with references to [1], the inputs we take as given: the analytic hypotheses on the model, the resolution package, the standard form of the log likelihood ratio, and the zeta function. A reader familiar with the main results of singular learning theory but not with the details of [1] should find here everything needed for the later sections.

2.1 The model and the fundamental conditions

Let $q(x)$ and $p(x \mid w)$, $w \in W \subset \mathbb{R}^d$, be probability densities on $\mathbb{R}^N$ with common support, W compact, and let

$$K(w) \;=\; \int q(x) \log \frac{q(x)}{p(x \mid w)}\, dx, \qquad W_0 \;=\; \{w \in W :\; K(w) = 0\} \neq \emptyset,$$

be the Kullback–Leibler function and the set of true parameters. Write $f(x, w) = \log(q(x)/p(x \mid w))$, so that $K(w) = \mathbb{E}_X[f(X, w)]$. Given n independent samples from q, the *log likelihood ratio function* is $K_n(w) = \frac{1}{n} \sum_{i=1}^{n} f(X_i, w)$, and the *normalized evidence* and *normalized free energy* at inverse temperature $\beta > 0$ are

$$Z_n^0 \;=\; \int_W e^{-n\beta K_n(w)}\,\varphi(w)\,dw, \qquad F_n^0 \;=\; -\log Z_n^0, \tag{2}$$

related to F_n by $F_n = n\beta S_n + F_n^0$, where $S_n = -\frac{1}{n}\sum_i \log q(X_i)$.

Assumption 2.1 (Fundamental conditions; [1, Def. 6.1, Def. 6.3])**.** *(I)* (index s, an even integer ≥ 2) The map $w \mapsto f(\cdot, w)$ extends to an $L^s(q)$-valued complex analytic function on an open set $W^{(\mathbb{C})} \subset \mathbb{C}^d$ containing W; the envelope $M(x) = \sup_{w\in W^{(\mathbb{C})}} |f(x,w)|$ lies in $L^s(q)$; and $\int M(x)^2 \sup_{K(w)\leq\epsilon} p(x \mid w)\,dx < \infty$ for some $\epsilon > 0$.

(II) $W = \{w : \pi_1(w) \geq 0, \ldots, \pi_k(w) \geq 0\}$ with π_j real analytic, and $\varphi = \varphi_1\varphi_2$ with $\varphi_1 > 0$ smooth and $\varphi_2 \geq 0$ real analytic.

Condition (I) at a larger even index implies it at any smaller one ([1, Rem. 6.1(7)]). Sections 3–6 use only $s = 2$, which is assumed by default throughout; Section 7 strengthens this to $s = 4$, stated there explicitly. Condition (I) is the analytic input driving the empirical process theory; condition (II) rules out pathological (flat but nonanalytic) priors and boundaries. We refer to [1, §6.1] for discussion.

2.2 The resolution package

The following consequence of Hironaka's theorem [2] is the geometric backbone; we fix its data once and for all.

Theorem 2.2 (Resolution package; [1, Thm. 6.5])**.** *Under Assumption 2.1 there exist $\epsilon \in (0,1)$, a d-dimensional real analytic manifold containing a compact subset M, and a proper real analytic map g from that manifold onto an open neighborhood of $W_\epsilon := \{w \in W : K(w) \leq \epsilon\}$, with $M = g^{-1}(W_\epsilon)$, such that M is covered by finitely many charts $U_\alpha \cong [0,b]^d$ (nonnegative orthants) in which:*

(1) $K_g := K \circ g$ is a normal crossing monomial, $K_g(u) = u^{2k} = u_1^{2k_1}\cdots u_d^{2k_d}$, with nonnegative integers $k_j = k_j^{(\alpha)}$;

(2) the pullback of the prior, $\omega(du) := \varphi(g(u))\,|g'(u)|\,du$, has the form $\omega(du) = \phi(u)\,u^h\,du$ with nonnegative integers $h_j = h_j^{(\alpha)}$ and $\phi \in C^\infty$, $\phi \geq c > 0$ on $[0,b]^d$;

(3) there is a smooth partition of unity $\{\sigma_\alpha\}$ with $\sum_\alpha \sigma_\alpha = 1$ on M, $\sigma_\alpha > 0$ on $[0,b)^d$ and $\operatorname{supp}\sigma_\alpha = [0,b]^d$; and for every integrable H,

$$\int_{W_\epsilon} H(w)\,\varphi(w)\,dw \;=\; \int_M H(g(u))\,\omega(du) \;=\; \sum_\alpha \int_{[0,b]^d} H(g(u))\,\sigma_\alpha(u)\,\phi(u)\,u^h\,du. \tag{3}$$

We emphasize that the measure ν on M and the function K_g are globally defined; only their normal crossing presentations are chart data.

2.3 The standard form and the process ξ_n

Theorem 2.3 (Standard form; [1, Thm. 6.1, Main Thm. 6.1])**.** *In each chart there is an $L^2(q)$-valued real analytic function $a(x,u)$ with*

$$f(x,g(u)) \;=\; a(x,u)\,u^k, \qquad \mathbb{E}_X[a(X,u)] = u^k.$$

Consequently, with

$$\xi_n(u) \;=\; \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\big(\mathbb{E}_X[a(X,u)] - a(X_i,u)\big), \tag{4}$$

the log likelihood ratio takes the standard form

$$K_n(g(u)) \;=\; K_g(u) \;-\; \frac{1}{\sqrt{n}}\sqrt{K_g(u)}\,\xi_n(u). \tag{5}$$

Three remarks. First, on signs: our ξ_n is the negative of [1, eq. (6.11)]; with the definition there, the standard form [1, eq. (6.10)] holds with a *plus* sign, so the two displays of [1] are mutually inconsistent as printed. The discrepancy is immaterial for results in law, since ξ and $-\xi$ are equal in law for a centred Gaussian process, but a definite convention matters once $a(x,u)$ enters expressions linearly (Section 8); ours agrees with that of ψ_n in (6) below. Second, on globality: since the charts are nonnegative orthants, $u^k \geq 0$, so u^k is the *nonnegative* square root $\sqrt{K_g(u)}$ in every chart; on overlaps the presentations therefore agree, and $a = f(\cdot,g(\cdot))/\sqrt{K_g}$ off the exceptional divisor, extended analytically across it, is a single global function on $\mathbb{R}^N \times M$. Hence ξ_n is a globally defined random element of $C(M)$, and (5) holds on all of M. Third, on the outer region: on $\{w : K(w) > \epsilon\}$ one uses instead the classical normalization

$$\psi_n(w) \;=\; \sqrt{n}\,\frac{K(w) - K_n(w)}{\sqrt{K(w)}}, \tag{6}$$

which is well defined there.

Theorem 2.4 ([1, Thms. 5.8, 5.9, 6.2, 6.3])**.** *Under Assumption 2.1, ξ_n converges in law in $C(M)$ (with the supremum norm) to a Gaussian process ξ with mean zero and covariance $\mathbb{E}[\xi(u)\xi(v)] = \mathbb{E}_X[a(X,u)a(X,v)] - u^k v^k$; moreover $\mathbb{E}[\xi(u)^2] = 2$ whenever $K_g(u) = 0$. Similarly ψ_n converges in law in $C(\{K \geq \epsilon\})$, and $\sup_n \mathbb{E}\big[\sup_M |\xi_n|^2\big] < \infty$, $\sup_n \mathbb{E}\big[\sup_{K\geq\epsilon} |\psi_n|^2\big] < \infty$.*

Two citation notes: in [1], the identity $\mathbb{E}[\xi(u)^2] = 2$ is [1, Thm. 6.3], stated there under index $s = 4$; its proof, however, uses only $M \in L^2(q)$ and condition (I) of Assumption 2.1, and is therefore available at $s = 2$. The uniform second-moment bounds are [1, Thm. 5.8] applied with $s = 2$.

2.4 The zeta function and the invariants (λ, m)

For $\Phi \in C(M)$ define

$$\zeta_\Phi(z) \;=\; \int_M K_g(u)^z\,\Phi(u)\,\omega(du). \tag{7}$$

Chartwise this is $\sum_\alpha \int u^{2kz+h}\,\Phi\,\sigma_\alpha\phi\,du$; the integral converges absolutely and is holomorphic for $\operatorname{Re} z > -\lambda$, where

$$\lambda \;=\; \min_{\alpha}\;\min_{j:\,k_j\geq 1}\;\frac{h_j+1}{2k_j}, \qquad m \;=\; \max_{\alpha:\;\min_j=\lambda}\;\#\Big\{j:\;\frac{h_j+1}{2k_j}=\lambda\Big\}. \tag{8}$$

(Coordinates with $k_j = 0$ do not constrain convergence and are assigned exponent $+\infty$.) By [1, Thm. 6.6], ζ_1 continues to a meromorphic function on $\mathbb{C}$ whose poles are negative rational and lie among the chartwise candidates in (8); that the largest pole is exactly $-\lambda$ and of order exactly m — i.e. that no cancellation occurs when the charts are summed — will follow from the positivity of the leading measure (Proposition 3.2(i)). Since $\int_W K^z\varphi\,dw - \zeta_1(z)$ is an integral over the compact set $\{K \geq \epsilon\}$, hence entire, (λ, m) agree with the invariants of the zeta function of the model on all of W; in particular they do not depend on the resolution.

Definition 2.5. A chart α is *essential* if the minimum in (8) equals λ there and is attained by exactly m indices. In an essential chart we relabel coordinates as $u = (x, y)$, $x = (u_1, \ldots, u_m)$ the *essential* directions (so $(h_j + 1)/(2k_j) = \lambda$, $k_j \geq 1$, for $j \leq m$) and $y = (u_{m+1}, \ldots, u_d)$ the rest, for which the inequality is strict:

$$\mu'_j \;:=\; h'_j - 2\lambda k'_j \;>\; -1 \qquad (m+1 \leq j \leq d), \tag{9}$$

writing (k', h') for the y-part of the multi-indices. Note that no chart contains more than m indices attaining λ, by maximality of m in (8).

3 The leading measure μ

Definition 3.1. For $\Phi \in C(M)$ set

$$\mu[\Phi] \;:=\; \lim_{\substack{\sigma\to-\lambda^+\\ \sigma\in\mathbb{R}}}\;(\sigma+\lambda)^m\,\zeta_\Phi(\sigma).$$

The definition involves no charts. Its existence, and everything we shall need, is:

Proposition 3.2. *(i) The limit exists for every $\Phi \in C(M)$ and defines a finite positive Radon measure μ on M with $0 < \mu(M) < \infty$. In particular $(\sigma+\lambda)^m\zeta_1(\sigma) \to \mu(M) > 0$, so the pole of ζ_1 at $-\lambda$ has order exactly m: positivity forbids cancellation across charts.*

(ii) In the notation of Definition 2.5, in each essential chart

$$d\mu \;=\; \frac{\sigma_\alpha(0,y)\,\phi(0,y)\,y^{\mu'}\,dy}{2^m\,k_1k_2\cdots k_m} \qquad \text{on the face } \{x=0\}, \tag{10}$$

summed over essential charts; non-essential charts contribute zero.

(iii) $\Sigma := \operatorname{supp}\mu$ is the union, over essential charts, of the closed faces $\{x = 0\}$. It is compact, of codimension m, contained in $g^{-1}(W_0)$, and admits the chart-free characterization: $u_0 \in \Sigma$ if and only if for every neighborhood $U \ni u_0$ there is $\Phi \geq 0$ in $C(M)$ supported in U with $\mu[\Phi] > 0$; equivalently, if and only if $(\sigma+\lambda)^m\zeta_\Phi(\sigma)$ has a strictly positive limit as $\sigma \to -\lambda^+$ for every $\Phi \geq 0$ continuous with $\Phi(u_0) > 0$ — a pole of order exactly m when Φ is smooth enough for ζ_Φ to continue meromorphically.

Proof. Write $\varepsilon = \sigma + \lambda > 0$. Fix a chart and put $\Psi = \Phi\,\sigma_\alpha\,\phi \in C([0,b]^d)$.

One-variable identity. For $\psi \in C[0,b]$ and $\kappa \geq 1$, substituting $v = (x/b)^{2\kappa\varepsilon}$,

$$2\kappa\varepsilon \int_0^b x^{2\kappa\varepsilon-1}\,\psi(x)\,dx \;=\; b^{2\kappa\varepsilon}\int_0^1 \psi\big(b\,v^{1/(2\kappa\varepsilon)}\big)\,dv \;\xrightarrow[\varepsilon\to 0^+]{}\; \psi(0) \tag{11}$$

by dominated convergence, since $v^{1/(2\kappa\varepsilon)} \to 0$ for each $v \in (0,1)$.

Essential charts. There $u^{2k\sigma+h} = \big(\prod_{j\leq m} x_j^{2k_j\varepsilon-1}\big)\big(\prod_{j>m} y_j^{\mu'_j+2k'_j\varepsilon}\big)$, using $h_j = 2k_j\lambda - 1$ for essential j and (9). Applying (11) in each essential variable (for fixed y) and dominated convergence in y — the y-integrand is dominated by $\|\Psi\|_\infty \max(y^{\mu'}, y^{\mu'+2k'})$, integrable by (9) — gives

$$\varepsilon^m \int_{[0,b]^d} u^{2k\sigma+h}\,\Psi(u)\,du \;\longrightarrow\; \frac{1}{2^m k_1\cdots k_m}\int_{[0,b]^{d-m}} \Psi(0,y)\,y^{\mu'}\,dy.$$

Non-essential charts. If the chart minimum λ_α exceeds λ, the chart integral is bounded near $\sigma = -\lambda$ (it converges absolutely for $\operatorname{Re} z > -\lambda_\alpha$), so $\varepsilon^m \cdot$ (chart integral) $\to 0$. If $\lambda_\alpha = \lambda$ with multiplicity $m_\alpha < m$, the computation just performed (with m_α in place of m) shows $\varepsilon^{m_\alpha} \cdot$ (chart integral) converges, hence again $\varepsilon^m \cdot$ (chart integral) $\to 0$.

Summing the finitely many charts proves existence of the limit and formula (10). Positivity of $\Phi \mapsto \mu[\Phi]$ is immediate from the definition, since $\zeta_\Phi(\sigma) \geq 0$ for $\Phi \geq 0$ and real $\sigma > -\lambda$; applying positivity to $\|\Phi\|_\infty \pm \Phi$ gives $|\mu[\Phi]| \leq \|\Phi\|_\infty\,\mu[1]$, and the Riesz representation theorem (or formula (10) directly) yields the measure. Since $\phi \geq c > 0$ and $\sigma_\alpha > 0$ on $[0,b)^d$, the density in (10) is strictly positive on a dense open subset of each essential face; hence $\mu(M) > 0$, each essential face lies in $\operatorname{supp}\mu$, and $\operatorname{supp}\mu$ is exactly the union of the closed essential faces. On such a face, $K_g = x^{2k}y^{2k'} = 0$ because every essential $k_j \geq 1$; thus $\Sigma \subseteq g^{-1}(W_0)$. The characterization in (iii) restates $u_0 \in \operatorname{supp}\mu$: the computation above shows that the one-sided limit exists, is finite, and equals $\mu[\Phi]$. □

Remark 3.3 (Dictionary with [1]). Watanabe's Theorem 4.6 computes, for tuned exponents $(h_j{+}1)/(2k_j) \equiv \lambda$ on $[0,b]^r$, the state density $v(t) = \int \delta(t - a\,x^{2k})\,x^h dx = \gamma_b\,t^{\lambda-1}a^{-\lambda}\big(\log(ab^{2|k|}/t)\big)^{r-1}$ with

$$\gamma_b \;=\; \frac{b^{\,|h|+r-2|k|\lambda}}{2^r\,(r-1)!\;k_1\cdots k_r}.$$

With $r = m$ and the exponents tuned, $|h| + r - 2|k|\lambda = 0$, so $(m-1)!\,\gamma_b = (2^m k_1\cdots k_m)^{-1}$, matching (10): chartwise, $d\mu = (m-1)!\,\gamma_b\,\sigma_\alpha(0,y)\phi(0,y)\,y^{\mu'}dy$. (The tuned case of this formula is rederived self-containedly as Lemma 4.4 below.) In the same spirit, an inverse Mellin transform ([1, Ex. 4.7]) converts Definition 3.1 into the leading term of the state density: for $\Phi \in C^\infty(M)$, the pushforward $(K_g)_*(\Phi\,\omega)$ has density

$$v_\Phi(t) \;=\; \mu[\Phi]\;\frac{t^{\lambda-1}(\log\frac{1}{t})^{m-1}}{(m-1)!}\,\big(1+o(1)\big), \qquad t \downarrow 0.$$

(For merely continuous Φ the asymptotic persists, but requires an equidistribution argument on the level sets rather than the exact Mellin pair; it is not used below.) Thus μ is precisely "the distribution formed by the leading coefficients of the expansion," and the consistency conditions among charts and partitions of unity that [1, Thm. 6.5] must arrange play no role: μ is defined without them.

Remark 3.4. For Φ smooth, ζ_Φ is meromorphic near $-\lambda$ (same continuation as [1, Thm. 6.6]) and $\mu[\Phi]$ is its top Laurent coefficient at $-\lambda$. For merely continuous Φ the continuation need not exist, but Definition 3.1 does not require it: the real one-sided limit is computed directly in Proposition 3.2. Positivity holds only for this top coefficient at the smallest exponent; the subleading coefficients are genuinely signed distributions.

4 Chart-level estimates

Fix $A \geq 0$ and $\beta > 0$ and set $w_A(t) := (1+t)^A\, e^{-\beta t/2}$ for $t \geq 0$. Call a measurable $G : [0, \infty) \to \mathbb{R}$ *admissible* if $|G| \leq w_A$. Constants C below depend only on (A, β) and the fixed resolution data, never on n, G, or the test functions. We write

$$r_n \ := \ \begin{cases} n^{-\lambda}\,(\log n)^{m-2}, & m \geq 2, \\ n^{-\lambda-\delta}\,(\log n)^{M_0}, & m = 1, \end{cases}$$

for constants $\delta > 0$ and $M_0 \leq d-1$ determined by the resolution and specified in the proof of Lemma 4.2.

Lemma 4.1 (A priori bound)**.** *For every admissible G and $\Phi \in C(M)$, and $n > e$,*

$$\Big|\int_M G\big(nK_g(u)\big)\,\Phi(u)\,\omega(du)\Big| \ \leq \ C\,\|\Phi\|_\infty\,\frac{(\log n)^{m-1}}{n^\lambda}.$$

More precisely, a single chart with data (k, h), exponent λ_α and multiplicity m_α contributes at most $C\,\|\Phi\|_\infty(\log n)^{m_\alpha-1}n^{-\lambda_\alpha}$.

Proof (cf. [1, Thms. 4.6–4.7]). By the partition of unity and $\|\sigma_\alpha\phi\|_\infty \leq C$ it suffices to prove the chart statement with $\Phi \equiv 1$, i.e. to bound $\int_{[0,b]^d} |G|(nu^{2k})\,u^h\,du$. Coordinates with $k_j = 0$ do not enter u^{2k} and integrate out into the constant (every chart contains an index with $k_j \geq 1$, since $K_g \leq \epsilon < 1$ on M); relabel the remaining coordinates so that $k_1, \ldots, k_{d'} \geq 1$ carry exponents $\lambda_1 \leq \cdots \leq \lambda_{d'}$, $\lambda_j = (h_j+1)/(2k_j)$, so that $\lambda_\alpha = \lambda_1$ and $m_\alpha = \#\{j : \lambda_j = \lambda_1\}$. Since $(1+t)^A e^{-\beta t/2} \leq C_{A,\beta}\,e^{-\beta t/4}$, we may and do take $G(t) = e^{-ct}$ with $c := \beta/4$.

Monomial substitution. With $v_j := u_j^{2k_j}$, so that $u_j^{h_j}\,du_j = \frac{1}{2k_j}\,v_j^{\lambda_j-1}\,dv_j$, and $B_j := b^{2k_j}$,

$$\int_{[0,b]^{d'}} e^{-cn\,u^{2k}}\,u^h\,du \ = \ \prod_{j\leq d'} \tfrac{1}{2k_j}\ \int_{\prod_j [0,B_j]} e^{-cn\,v_1\cdots v_{d'}}\ \prod_{j\leq d'} v_j^{\lambda_j-1}\,dv.$$

The chart data has been reduced to its list of exponents; the logarithms will be produced by ties at the minimum.

A folding bound. For $\lambda > 0$ and an integer $\nu \geq 0$ put $g_{\lambda,\nu}(s) := \min(1, s^{-\lambda})\,(1+\log_+ s)^\nu$, $s > 0$. We claim that for $\mu > 0$ and $B > 0$ there is C with

$$\int_0^B g_{\lambda,\nu}(sv)\,v^{\mu-1}\,dv \ \leq \ C\,g_{\lambda',\nu'}(s) \quad \text{for all } s > 0, \qquad (\lambda', \nu') = \begin{cases} (\mu,\,0), & \mu < \lambda, \\ (\lambda,\,\nu+1), & \mu = \lambda, \\ (\lambda,\,\nu), & \mu > \lambda. \end{cases} \tag{12}$$

Indeed, for $s \le 1$ the integrand is at most $(1+\log_+ B)^\nu v^{\mu-1}$ and both sides are of order one. For $s>1$, split at $v=1/s$: the piece $v \le \min(B,1/s)$ contributes at most $s^{-\mu}/\mu$; if $B>1/s$, the substitution $w=sv$ turns the remaining piece into

$$s^{-\mu}\int_1^{sB} w^{\mu-\lambda-1}\,(1+\log w)^\nu\,dw,$$

whose integral is bounded by a constant if $\mu<\lambda$, by $C(1+\log sB)^{\nu+1}$ if $\mu=\lambda$, and by $C(sB)^{\mu-\lambda}(1+\log sB)^\nu$ if $\mu>\lambda$; using $1+\log_+(sB) \le (1+\log_+ B)(1+\log s)$ for $s>1$, each case matches (12).

Iteration. For every $s>0$, $\int_0^{B_{d'}} e^{-sv}v^{\lambda_{d'}-1}dv \le \min\bigl(B_{d'}^{\lambda_{d'}}/\lambda_{d'},\ \Gamma(\lambda_{d'})\,s^{-\lambda_{d'}}\bigr) \le C\,g_{\lambda_{d'},0}(s)$. Now integrate the coordinates in the order $v_{d'}, v_{d'-1},\dots,v_1$ — largest exponents first — applying (12) at each step with s equal to cn times the product of the not-yet-integrated coordinates. The running parameters evolve by the rule in (12): an exponent equal to the current minimum increments the logarithm count, an exponent strictly smaller resets it to zero. Since the exponents are processed in decreasing order, the last reset happens when the value λ_1 is first met (at index m_α), and the remaining $m_\alpha-1$ steps are ties; hence the final parameters are $(\lambda_1,\ m_\alpha-1)$, ties at non-minimal values having been wiped out by the reset. Therefore

$$\int_{[0,b]^{d'}} e^{-cn\,u^{2k}}\,u^h\,du \ \le\ C\,g_{\lambda_1,\,m_\alpha-1}(cn) \ \le\ C'\,\frac{(\log n)^{m_\alpha-1}}{n^{\lambda_\alpha}} \qquad (n>e). \qquad \square$$

Two conventions govern how Lemma 4.1 is applied below. First, its conclusion is homogeneous in G: if $|G| \le N\,w_A$ pointwise for some constant N, the bound holds with the extra factor N (apply the lemma to G/N). Second, an integrand depending on further variables besides t — an insertion $F(t,u)$, say — is fed to the lemma through its *envelope*: from a bound of the form $\sup_{v\in M}|F(t,v)| \le N\,w_A(t)$ one obtains $|F(nK_g(u),u)| \le N\,w_A(nK_g(u))$ pointwise in u, and the lemma is then applied to the admissible function w_A itself, at the price of the factor N. Note that neither step fixes the second argument: for fixed u the function $t\mapsto F(t,u)$ need not be admissible with constant one, and in the integrals below the second argument moves with the integration variable in any case; it is discarded at the level of absolute values. Both conventions are used freely in what follows.

Lemma 4.2 (Vanishing estimate)**.** *Let $\Psi:[0,\infty)\times M\to\mathbb{R}$ be jointly measurable with $\Psi(t,\cdot)\in C^1(M)$ for each t,*

$$N_A(\Psi)\ :=\ \sup_{t,u}\ \frac{|\Psi(t,u)|+|\nabla_u\Psi(t,u)|}{w_A(t)}\ <\ \infty,$$

and suppose $\Psi(t,u)=0$ for all $t\ge 0$ and all $u\in\Sigma$. Then for $n>e$,

$$\Bigl|\int_M \Psi\bigl(nK_g(u),u\bigr)\,\omega(du)\Bigr|\ \le\ C\,N_A(\Psi)\,r_n.$$

(Here ∇_u and the C^1 norm are taken with respect to the fixed finite atlas; equivalently, any fixed Riemannian metric.)

Proof. Decompose by the partition of unity and treat charts separately.

Essential charts. The face $\{x=0\}$ lies in Σ by Proposition 3.2(iii), so $\Psi(t,(0,y))=0$. The chart $[0,b]^d$ is convex, so the segment from (x,y) to $(0,y)$ stays in the chart, and the mean value theorem gives

$$|\Psi(t,(x,y))| \;=\; |\Psi(t,(x,y))-\Psi(t,(0,y))| \;\le\; N_A(\Psi)\, w_A(t) \sum_{j\le m} x_j.$$

Hence the chart integral is bounded by $C\,N_A(\Psi)\sum_{j\le m}\int_{[0,b]^d} w_A(nu^{2k})\,u^{h+e_j}\,du$, where e_j denotes the j-th standard unit multi-index (so $u^{h+e_j}=u^h\,u_j$). The shifted data $(k,\,h+e_j)$ has j-th exponent $\lambda+\frac{1}{2k_j}$ and all other exponents unchanged. If $m\ge 2$, the minimum is still λ, now of multiplicity $m-1$; Lemma 4.1 (chart version) gives the bound $C\,N_A(\Psi)\,(\log n)^{m-2}n^{-\lambda}$. If $m=1$, the new minimum is at least $\lambda+\delta_\alpha$ with $\delta_\alpha := \min\big(\frac{1}{2k_j},\ \min_{i>m}\frac{h_i'+1}{2k_i'}-\lambda\big)>0$, and the chart bound is $C\,N_A(\Psi)\,(\log n)^{d-1}n^{-\lambda-\delta_\alpha}$.

Non-essential charts. No vanishing is needed: Lemma 4.1 gives $(\log n)^{m_\alpha-1}n^{-\lambda_\alpha}$ with either $\lambda_\alpha=\lambda,\ m_\alpha\le m-1$ (absorbed into $(\log n)^{m-2}n^{-\lambda}$) or $\lambda_\alpha>\lambda$ (absorbed into either rate, since $n^{-(\lambda_\alpha-\lambda)}$ beats any power of $\log n$; note $m=1$ forces $\lambda_\alpha>\lambda$ in every non-essential chart).

Take $\delta := \min\Big(\ \min_\alpha \delta_\alpha,\ \ \min\{\lambda_\alpha-\lambda:\ \lambda_\alpha>\lambda\}\ \Big)$ and $M_0 := d-1$. □

Remark 4.3. Lemma 4.2 is the precise sense in which "the coefficient distributions one order below the top have order at most one": subtracting the value on Σ costs one derivative of the test function and gains one power of $\log n$ (or, when $m=1$, a power $n^{-\delta}$). For $\Psi(t,u)=G(t)\Phi(u)$ with $\Phi\in C^\infty$ vanishing on Σ, the same chart computation shows that ζ_Φ has a pole of order at most $m-1$ at $-\lambda$.

Finally, the exact evaluation that powers the kernel theorem. It concerns the *tuned* case, in which all exponents of a monomial block agree — precisely the configuration of the essential directions — and in that case the pushforward to the t-variable has a closed form: the same substitution $v_j=x_j^{2k_j}$ as above reduces it, in logarithmic coordinates, to the elementary fact that the sum of the coordinates of $[0,\infty)^m$ carries the Gamma density. This is where the $(\log)^{m-1}$ originates.

Lemma 4.4 (Tuned state density; cf. [1, Thm. 4.6])**.** *Let $m\ge 1$, let $k_1,\dots,k_m\ge 1$ be integers, $b>0$, $\lambda>0$, and suppose the exponents are tuned: $h_j=2k_j\lambda-1$ for $j\le m$. For every $a>0$, the pushforward of the measure $x^h\,dx$ on $[0,b]^m$ under $x\mapsto \tau=a\,x^{2k}$ is carried by $(0,\,ab^{2|k|})$, $|k|:=k_1+\cdots+k_m$, with density*

$$v_a(\tau) \;=\; \frac{1}{2^m\,k_1\cdots k_m\,(m-1)!}\;\frac{\tau^{\lambda-1}}{a^\lambda}\;\Big(\log\frac{a\,b^{2|k|}}{\tau}\Big)^{m-1}.$$

Proof. The substitution $v_j=x_j^{2k_j}$ of Lemma 4.1 replaces $x^h\,dx$ by $\prod_j \frac{1}{2k_j}\,v_j^{\lambda-1}dv_j$ on $\prod_j[0,B_j]$, $B_j=b^{2k_j}$, with $\tau=a\,v_1\cdots v_m$; all exponents are now equal to λ. Pass to logarithmic coordinates $w_j:=\log(B_j/v_j)\in[0,\infty)$: the measure becomes $\prod_j\frac{B_j^\lambda}{2k_j}\,e^{-\lambda w_j}\,dw_j$, while $\tau=a\,b^{2|k|}e^{-W}$ with $W:=\sum_j w_j$. The pushforward of Lebesgue measure on $[0,\infty)^m$ under $w\mapsto W$ has density $W^{m-1}/(m-1)!$ (differentiate $\mathrm{vol}\{w\ge 0:\sum_j w_j\le W\}=W^m/m!$), and the factor $e^{-\lambda\sum_j w_j}=e^{-\lambda W}$ is constant on the level sets; so the pushforward of the measure under W has density $\frac{W^{m-1}}{(m-1)!}e^{-\lambda W}$ times $\prod_j\frac{B_j^\lambda}{2k_j}$. Finally $W=\log(ab^{2|k|}/\tau)$, $dW=d\tau/\tau$, and

$e^{-\lambda W} = \bigl(\tau/(ab^{2|k|})\bigr)^{\lambda}$, whence

$$v_a(\tau) \;=\; \prod_j \frac{B_j^{\lambda}}{2k_j} \cdot \frac{1}{(m-1)!} \Bigl(\log \frac{ab^{2|k|}}{\tau}\Bigr)^{m-1} \Bigl(\frac{\tau}{ab^{2|k|}}\Bigr)^{\lambda} \frac{1}{\tau},$$

which is the stated formula, the powers of b cancelling by $\prod_j B_j^{\lambda} = b^{2|k|\lambda}$. $\square$

In particular $(m-1)!\, v_a$ has leading constant $(2^m k_1 \cdots k_m)^{-1}$, identifying the constant γ_b of Remark 3.3 in the tuned case.

5 The kernel theorem

Definition 5.1. For jointly measurable $F : [0,\infty) \times M \to \mathbb{R}$ with $F(t,\cdot) \in C^1(M)$ for each t and $N_A(F) < \infty$, define

$$\Lambda_n[F] \;:=\; \frac{(m-1)!\, n^{\lambda}}{(\log n)^{m-1}} \int_M F\bigl(nK_g(u),\, u\bigr)\, \omega(du), \qquad \Lambda[F] \;:=\; \int_0^{\infty}\!\! \int_{\Sigma} F(t,u)\, t^{\lambda-1}\, d\mu(u)\, dt.$$

Theorem 5.2 (Kernel theorem)**.** *There exist C and, when $m=1$, constants $\delta' > 0$, $M_0' \geq 0$, depending only on (A,β) and the resolution, such that for all such F and all $n > e$,*

$$\bigl|\Lambda_n[F] - \Lambda[F]\bigr| \;\leq\; C\, N_A(F) \times \begin{cases} (\log n)^{-1}, & m \geq 2, \\ n^{-\delta'} (\log n)^{M_0'}, & m = 1. \end{cases}$$

Proof. The shape of the argument. Write $\Lambda_n[F] = c_n \sum_\alpha I_n^\alpha$ with $c_n := \frac{(m-1)!\, n^{\lambda}}{(\log n)^{m-1}}$ and $I_n^\alpha = \int_{[0,b]^d} F(nu^{2k}, u)\, \sigma_\alpha \phi\, u^h\, du$ (we suppress the chart identification in the second argument of F). Non-essential charts contribute only error terms (Step 0). In an essential chart, two things separate Λ_n from Λ. First, the integrand depends on u through more than the pair $(t,y) = (nK_g(u), y)$: both F and the smooth density $\sigma_\alpha\phi$ vary in the essential directions x, whereas the limit sees only their values on the face $\{x=0\}$. Step 1 removes this by restricting the integrand to the face, at the cost of one power of $\log n$ (or $n^{-\delta}$) — the vanishing mechanism of Lemma 4.2; the restricted integral is called J_n^α. Second, in J_n^α the x-dependence is purely monomial and *tuned*, so for each fixed y the x-integral is computed *exactly* by Lemma 4.4 with scale $a = n\, y^{2k'}$: it is the integral of $F(t,(0,y))$ against an explicit density on $(0, T_y)$, where

$$T_y \;:=\; n\, y^{2k'} b^{2|k|}$$

is the largest value that $t = nK_g$ can attain over the chart at that y. The density carries the factor $\bigl(\log(T_y/t)\bigr)^{m-1} = \bigl(\log n + c(t,y)\bigr)^{m-1}$, which Step 2 expands in powers of $\log n$: the top power reproduces the chart's share of $\Lambda[F]$ *except* that t runs only up to T_y, not to ∞. The missing region $\{t \geq T_y\}$ is the truncation error, small because $t \geq T_y$ forces $y^{2k'} \leq t/(nb^{2|k|})$, a set of small y-measure (Step 3); and the lower powers of $\log n$ are one order down (Step 4). Step 5 assembles the charts into the statement.

Step 0: non-essential charts. Pointwise in u, $|F(nu^{2k}, u)| \leq \sup_{v\in M} |F(nu^{2k}, v)| \leq N_A(F)\, w_A(nu^{2k})$; applying Lemma 4.1 to the admissible envelope w_A itself (chart version, $\Phi \equiv 1$) gives $|I_n^\alpha| \leq$

$C\,N_A(F)\,(\log n)^{m_\alpha-1}n^{-\lambda_\alpha}$, which after normalization is $O\big(N_A(F)(\log n)^{-1}\big)$, resp. $O(N_A(F)\,n^{-\delta'}(\log n)^{M_0'})$ when $m=1$, as in the proof of Lemma 4.2.

Step 1: restriction to the face. Set

$$J_n^\alpha \;:=\; \int_{[0,b]^d} F\big(nx^{2k}y^{2k'},\,(0,y)\big)\,\sigma_\alpha(0,y)\,\phi(0,y)\;x^hy^{h'}\,dx\,dy.$$

Only the *amplitude* has been restricted — the dependence of F on the point u, and the smooth density $\sigma_\alpha\phi$ — and the selection is by modulus of continuity, not by occurrences of x. The restricted ingredients are C^1 in u with n-independent norms, so replacing them by their face values is priced by the vanishing mechanism below. The two ingredients kept exact are of a different nature. The first argument $t=nx^{2k}y^{2k'}$ is the fast variable whose distribution is the object of the limit: as a function of u its derivative is of order n, and freezing it would alter the integrand by $O(1)$ precisely on the region $\{nK_g\asymp 1\}$ carrying the mass. The monomial $x^hy^{h'}\,dx\,dy$ is the reference measure: it has no trace on the face, where it vanishes identically, and it reaches the limit by *pushforward* — Lemma 4.4 — rather than by restriction. This is the division of the classical Laplace method (freeze the amplitude at the critical locus; never the phase or the volume element), and it restricts exactly as much as Step 2 needs: in J_n^α the variables x enter only through t and through the reference measure, so the x-integral becomes a monomial pushforward.

The integrand of $I_n^\alpha - J_n^\alpha$ is $\Psi(nu^{2k},u)\,u^h$ where $\Psi(t,(x,y)) := F(t,(x,y))\sigma_\alpha(x,y)\phi(x,y) - F(t,(0,y))\sigma_\alpha(0,y)\phi(0,y)$ vanishes at $x=0$ and satisfies $|\nabla_x\Psi|\le C\,N_A(F)\,w_A(t)$ (product rule, using $\|\sigma_\alpha\phi\|_{C^1}\le C$). Exactly as in the proof of Lemma 4.2 (mean value theorem in the convex chart, then Lemma 4.1 with data $(k,h+e_j)$),

$$|I_n^\alpha - J_n^\alpha| \;\le\; C\,N_A(F)\,r_n.$$

Step 2: exact evaluation of J_n^α. Fix y with $y^{2k'}>0$ (the complement is ω-null on the chart). By Lemma 4.4, applied to the tuned block x with scale $a=n\,y^{2k'}$, the pushforward of $x^h\,dx$ on $[0,b]^m$ under $x\mapsto t=n\,y^{2k'}x^{2k}$ is

$$\gamma_b\,t^{\lambda-1}\,\big(n\,y^{2k'}\big)^{-\lambda}\Big(\log\frac{n\,y^{2k'}b^{2|k|}}{t}\Big)^{m-1}dt \qquad \text{on } \big(0,\;T_y\big), \qquad \gamma_b=\frac{1}{2^mk_1\cdots k_m\,(m-1)!}\,.$$

Hence, by Fubini, and writing $c(t,y):=\log\big(y^{2k'}b^{2|k|}/t\big)$,

$$J_n^\alpha \;=\; \gamma_b\,n^{-\lambda}\int dy\;\sigma_\alpha(0,y)\phi(0,y)\,y^{\mu'}\int_0^{T_y}F\big(t,(0,y)\big)\,t^{\lambda-1}\,\big(\log n + c(t,y)\big)^{m-1}\,dt,$$

where $y^{h'}(y^{2k'})^{-\lambda}=y^{\mu'}$. Expand the binomial $(\log n+c)^{m-1}=\sum_{j=0}^{m-1}\binom{m-1}{j}(\log n)^{m-1-j}c^j$. The $j=0$ term equals, after multiplying by the normalization c_n and recalling $(m-1)!\,\gamma_b\,\sigma_\alpha(0,y)\phi(0,y)y^{\mu'}dy = d\mu|_\alpha$ from (10),

$$\Lambda^\alpha[F] \;-\; \underbrace{\iint_{t\ge T_y}F\,t^{\lambda-1}\,d\mu|_\alpha\,dt}_{\text{truncation}}, \qquad \Lambda^\alpha[F]:=\int_0^\infty\!\!\int F(t,u)\,t^{\lambda-1}\,d\mu|_\alpha(u)\,dt.$$

Step 3: the truncation. Note $T_y \le t$ iff $y^{2k'} \le s := t/(nb^{2|k|})$. Choose δ_1 (depending on the chart) with $0 < \delta_1 < \min\{(\mu'_j + 1)/(2k'_j) : k'_j \ge 1\}$ (any $\delta_1 > 0$ if no $k'_j \ge 1$). The pointwise Chebyshev bound $\mathbf{1}\{y^{2k'} \le s\} \le (s/y^{2k'})^{\delta_1}$, valid off the null set $\{y^{2k'} = 0\}$, gives

$$\int_{\{y^{2k'} \le s\}} y^{\mu'}\, dy \;\le\; s^{\delta_1} \int_{[0,b]^{d-m}} y^{\mu' - 2\delta_1 k'}\, dy \;=\; C\, s^{\delta_1} \qquad (s > 0),$$

the last integral being finite by the choice of δ_1 and (9). Hence, using $|F(t,u)| \le N_A(F)\, w_A(t)$ and Fubini, the truncation is at most $C\, N_A(F)\, n^{-\delta_1} \int_0^\infty w_A(t)\, t^{\lambda+\delta_1-1}\, dt \le C\, N_A(F)\, n^{-\delta_1}$. (At the corresponding point the proof of [1, Thm. 4.9] cites [1, Cor. 4.3], which is stated only for tuned exponents; the display above supplies the untuned case, without the logarithmic factor.)

Step 4: subleading logarithms. The terms $1 \le j \le m-1$ are bounded by

$$\binom{m-1}{j} (\log n)^{m-1-j}\, \gamma_b\, n^{-\lambda}\, N_A(F) \iint w_A(t)\, t^{\lambda-1}\, |c(t,y)|^j\, y^{\mu'}\, dt\, dy,$$

and the double integral is finite because $\lambda > 0$, $\mu'_j > -1$, and w_A decays exponentially; after normalization these are $O(N_A(F)(\log n)^{-1})$.

Step 5: assembly. Steps 2–4 give, for each essential chart,

$$\left| c_n\, J_n^\alpha - \Lambda^\alpha[F] \right| \;\le\; C\, N_A(F) \left(n^{-\delta_1} + (\log n)^{-1}\right)$$

(for $m = 1$ there are no subleading terms and the bound is $C\, N_A(F)\, n^{-\delta_1}$). Since $\sum_\alpha \sigma_\alpha \equiv 1$ makes the chart decomposition $\Lambda_n[F] = c_n \sum_\alpha I_n^\alpha$ exact, the triangle inequality over the finitely many charts yields

$$\Big| \Lambda_n[F] - \sum_{\alpha \text{ essential}} \Lambda^\alpha[F] \Big| \;\le\; \sum_{\alpha \text{ non-essential}} c_n \left| I_n^\alpha \right| \;+\; \sum_{\alpha \text{ essential}} \Big(c_n \left| I_n^\alpha - J_n^\alpha \right| + \left| c_n\, J_n^\alpha - \Lambda^\alpha[F] \right| \Big),$$

and every summand is within the stated rates, by Step 0, Step 1, and the display above; when $m = 1$, take $\delta' := \min\big(\delta,\ \min_\alpha \delta_1\big)$, the inner minimum over the essential charts, and $M'_0 := M_0$, where δ and M_0 are the constants of the rate r_n, specified in the proof of Lemma 4.2 and entering here through Steps 0 and 1. It remains to observe that

$$\sum_{\alpha \text{ essential}} \Lambda^\alpha[F] \;=\; \Lambda[F],$$

and this is an *identity*, not an estimate: by Proposition 3.2(ii), the measure μ *is* the sum, over the essential charts, of the pieces $d\mu|_\alpha$, non-essential charts contributing zero — because the chart-free limit of Definition 3.1 was evaluated, in the proof of Proposition 3.2, through the same partition of unity used here. Overlapping charts share their mass in the proportions σ_α in both computations, so no compatibility between charts needs to be checked; this is the point at which the chart-free definition of μ replaces the consistency conditions of [1, Thm. 6.5(3)–(4)]. □

Corollary 5.3 (All insertions of Watanabe type)**.** *For $F(t,u) = t^p e^{-\beta t} \Phi(u)$ with $p \ge 0$ and $\Phi \in C^1(M)$,*

$$\Lambda_n[F] \;\longrightarrow\; \frac{\Gamma(\lambda + p)}{\beta^{\lambda+p}}\, \mu[\Phi].$$

Remark 5.4 (On the exponent conventions). The exponent of n in Λ_n is λ for *every* p: Corollary 5.3 inserts the renormalized quantity $t^p = (nK_g)^p$. Watanabe's Theorem 4.7 inserts $K_g^p = (t/n)^p$ instead, producing the rate $n^{-(\lambda+p)}$; the two statements differ by the explicit factor n^{-p} and are equivalent. The renormalized convention is the one consumed downstream (posterior moments of nK_n, fluctuation functionals), which is why Theorem 5.2 is stated for general $F(t,u)$: nothing further will be needed from singular integral theory.

6 The limit law of the evidence and the free energy formula

Lemma 6.1 (Gradient moments). $\sup_n \ \mathbb{E}\Big[\|\xi_n\|^2_{C^1(M)}\Big] \ < \ \infty$, *where* $\|\eta\|_{C^1(M)} := \sup_M |\eta| + \max_\alpha \max_j \ \sup_{U_\alpha} |\partial_{u_j}\eta|$.

Proof. The bound for $\sup_M |\xi_n|$ is Theorem 2.4. For the gradient: on (a complex neighborhood of) each closed chart, $u \mapsto a(\cdot,u)$ is $L^2(q)$-valued analytic (Theorem 2.3), with Taylor coefficients $a_\gamma \in L^2(q)$ obeying the Cauchy estimates $\|a_\gamma\|_2 \le \|M_a\|_2/R^\gamma$ of [1, Rem. 5.5, eq. (5.11)]. Term-by-term differentiation then exhibits $u \mapsto \partial_{u_j} a(\cdot,u)$ as an $L^2(q)$-valued analytic family on a slightly smaller polydisc, with summable coefficient norms. Since $\partial_{u_j}\xi_n$ is, up to sign, the centred empirical process of this family — differentiation under $\mathbb{E}_X$ being justified by the same estimates — [1, Thm. 5.8] with $s = 2$ gives $\mathbb{E}[\sup_{U_\alpha} |\partial_{u_j}\xi_n|^2] \le C_{\alpha,j}$, uniformly in n. Sum over the finite atlas. $\square$

Definition 6.2. For $\eta \in C(M)$ set

$$\mathcal{Y}(\eta) \ := \ \frac{1}{(m-1)!} \int_0^\infty \int_\Sigma t^{\lambda-1}\, e^{-\beta t + \beta\sqrt{t}\,\eta(u)}\, d\mu(u)\, dt.$$

Lemma 6.3. $0 < \mathcal{Y}(\eta) < \infty$ *for every* $\eta \in C(M)$, *and* $\mathcal{Y}$ *is locally Lipschitz for the supremum norm: for* $\|\eta\|_\infty, \|\eta'\|_\infty \le R$,

$$|\mathcal{Y}(\eta) - \mathcal{Y}(\eta')| \ \le \ \frac{\beta\,\mu(M)\, e^{\beta R^2/2}}{(m-1)!} \Big(\int_0^\infty t^{\lambda-\frac{1}{2}} e^{-\beta t/2}\, dt\Big)\, \|\eta - \eta'\|_\infty.$$

Proof. By $\beta\sqrt{t}\,\eta \le \beta t/2 + \beta\eta^2/2$, the integrand is at most $t^{\lambda-1}e^{-\beta t/2}e^{\beta R^2/2}$, giving finiteness; positivity holds because the integrand is strictly positive and $\mu(M) > 0$ (Proposition 3.2). The Lipschitz bound follows from $|e^a - e^{a'}| \le e^{\max(a,a')}|a - a'|$ with $a = \beta\sqrt{t}\,\eta(u)$, absorbing $e^{\beta\sqrt{t}R} \le e^{\beta t/2 + \beta R^2/2}$. $\square$

Theorem 6.4 (Limit law of the normalized evidence). *Under Assumption 2.1,*

$$\frac{n^\lambda}{(\log n)^{m-1}}\, Z_n^0 \ \longrightarrow \ \mathcal{Y}(\xi) \qquad \textit{in law.}$$

Proof. Split $Z_n^0 = Z_n^{(1)} + Z_n^{(2)}$ over W_ϵ and $\{K > \epsilon\}$.

Outer region. On $\{K > \epsilon\}$, by (6) and $2\sqrt{nK}\,\psi_n \le nK + \psi_n^2$,

$$n\beta K_n \ = \ n\beta K - \beta\sqrt{nK}\,\psi_n \ \ge \ \tfrac{\beta}{2}\big(n\epsilon - \sup_{K>\epsilon} \psi_n^2\big),$$

so $\frac{n^\lambda}{(\log n)^{m-1}} Z_n^{(2)} \le \frac{n^\lambda e^{-n\beta\epsilon/2}}{(\log n)^{m-1}} \exp\big(\frac{\beta}{2}\sup_{K>\epsilon}\psi_n^2\big)$, which tends to 0 in probability since $\sup_{K>\epsilon}|\psi_n|$ converges in law (Theorem 2.4); this is display (6.18) of [1] and is already coordinate-free.

Inner region. By (3) and the standard form (5), using $\sqrt{n}\sqrt{K_g(u)} = \sqrt{nK_g(u)}$,

$$Z_n^{(1)} \;=\; \int_M \exp\big(-n\beta K_g(u) + \beta\sqrt{nK_g(u)}\,\xi_n(u)\big)\,\omega(du) \;=\; \int_M F_n\big(nK_g(u),u\big)\,\omega(du),$$

where $F_n(t,u) := e^{-\beta t + \beta\sqrt{t}\,\xi_n(u)}$. Thus $\frac{n^\lambda}{(\log n)^{m-1}} Z_n^{(1)} = \frac{1}{(m-1)!}\Lambda_n[F_n]$. The bounds $\beta\sqrt{t}\,\xi_n \le \beta t/2 + \beta\xi_n^2/2$ and $|\nabla_u F_n| = \beta\sqrt{t}\,|\nabla_u \xi_n|\,F_n$ give, with $A = \frac{1}{2}$,

$$N_{1/2}(F_n) \;\le\; C_\beta\,\big(1 + \beta\|\xi_n\|_{C^1}\big)\,e^{\beta\|\xi_n\|_\infty^2/2},$$

which is bounded in probability by Lemma 6.1 and Markov's inequality. Theorem 5.2 therefore yields

$$\Big|\frac{n^\lambda\, Z_n^{(1)}}{(\log n)^{m-1}} - \mathcal{Y}(\xi_n)\Big| \;=\; \frac{\big|\Lambda_n[F_n] - \Lambda[F_n]\big|}{(m-1)!} \;\le\; C\,N_{1/2}(F_n)\cdot o(1) \;\longrightarrow\; 0 \quad \text{in probability,}$$

noting $\Lambda[F_n]/(m-1)! = \mathcal{Y}(\xi_n)$ by Definitions 5.1 and 6.2. Finally $\xi_n \to \xi$ in law in $C(M)$ (Theorem 2.4) and $\mathcal{Y}$ is continuous (Lemma 6.3), so $\mathcal{Y}(\xi_n) \to \mathcal{Y}(\xi)$ in law by the continuous mapping theorem [1, Thm. 5.1]; conclude by Slutsky's theorem [1, Thm. 5.2(2)]. □

Corollary 6.5 (Free energy formula; cf. [1, Main Thm. 6.2(1), Cor. 6.1(1)])**.** *In law,*

$$F_n^0 - \lambda\log n + (m-1)\log\log n \;\longrightarrow\; -\log\mathcal{Y}(\xi),$$

and consequently

$$F_n \;=\; n\beta S_n + \lambda\log n - (m-1)\log\log n + F_n^R, \qquad F_n^R \;\longrightarrow\; -\log\mathcal{Y}(\xi) \text{ in law.}$$

Proof. $\mathcal{Y}(\xi) \in (0,\infty)$ almost surely (Lemma 6.3 applied to the continuous paths of ξ), and $-\log$ is continuous on $(0,\infty)$; apply the continuous mapping theorem to Theorem 6.4, and use $F_n = n\beta S_n + F_n^0$. □

No essential family of coordinates, no chart constant γ_b, and no partition of unity appears in Theorem 6.4 or Corollary 6.5: the limit is a functional of the pair $(\mu,\ \xi|_\Sigma)$ together with the invariants (λ, m). (The pair depends on the chosen resolution; the law of $\mathcal{Y}(\xi)$ does not — see Remark 9.1.)

7 Convergence of expectations

In this section Assumption 2.1(I) is strengthened to index $s = 4$; all other notation and standing data are as before. The goal is the expectation version of Corollary 6.5, i.e. the coordinate-free form of [1, Main Thm. 6.2(2), Cor. 6.1(2)]. Throughout, write

$$A_n \;:=\; F_n^0 - \lambda\log n + (m-1)\log\log n \;=\; -\log\Big(\frac{n^\lambda}{(\log n)^{m-1}}\,Z_n^0\Big) \qquad (n \ge 3),$$

so that Corollary 6.5 states $A_n \to -\log\mathcal{Y}(\xi)$ in law.

7.1 Background: asymptotic uniform integrability

Definition 7.1 ([1, Def. 5.2])**.** A sequence of real random variables $\{X_n\}$ is *asymptotically uniformly integrable* (AUI) if

$$\lim_{M\to\infty} \limsup_{n\to\infty} \sup_{N\ge n} \mathbb{E}\big[\, |X_N|\, \mathbf{1}\{|X_N| \ge M\} \,\big] \;=\; 0.$$

Theorem 7.2 ([1, Thms. 5.3–5.5])**.** *(a) If $|X_n| \le Y_n$ with $\{Y_n\}$ AUI, then $\{X_n\}$ is AUI; the property is preserved under multiplication by a constant (immediate from Definition 7.1).*

(b) If $\sup_n \mathbb{E}[X_n^2] < \infty$, then $\{X_n\}$ is AUI ([1, Thm. 5.4(3)] with $s = 2$, $\delta = 1$).

(c) If $X_n \to X$ in law, $\sup_n \mathbb{E}[|X_n|] < \infty$, and $\{X_n\}$ is AUI, then $\mathbb{E}[|X|] < \infty$ and $\mathbb{E}[X_n] \to \mathbb{E}[X]$.

7.2 Fourth moments

Lemma 7.3. *Under Assumption 2.1 with index $s = 4$,*

$$\sup_n \; \mathbb{E}\Big[\sup_M |\xi_n|^4\Big] \;<\; \infty, \qquad \sup_n \; \mathbb{E}\Big[\sup_{\{K\ge\epsilon\}} |\psi_n|^4\Big] \;<\; \infty.$$

Proof. Under (I) at index $s = 4$, the function $a(\cdot, u)$ of Theorem 2.3 is $L^4(q)$-valued analytic on each chart ([1, Thm. 6.1] applied at index 4), and the family $w \mapsto \big(K(w) - f(\cdot, w)\big)/\sqrt{K(w)}$ defining ψ_n is $L^4(q)$-valued analytic on a neighborhood of the compact set $\{K \ge \epsilon\}$, since $K \ge \epsilon > 0$ there (cf. the proof of [1, Thm. 6.2]). Apply [1, Thm. 5.8] with $s = 4$ on the finite atlas, resp. on a finite cover of $\{K \ge \epsilon\}$. Only these uniform bounds are needed below; the convergence of the corresponding moments ([1, Thm. 6.2(2)]) is not used. □

7.3 A two-sided envelope for A_n

Set $V_n := \max\big(\|\xi_n\|_\infty^2,\ \sup_{\{K\ge\epsilon\}} \psi_n^2\big)$.

Lemma 7.4 (Envelope)**.** *There exist a deterministic n_0 and a constant C_0 such that, surely, for all $n \ge n_0$:*

(i) $A_n \;\le\; \dfrac{\beta}{2}\, \|\xi_n\|_\infty^2 + C_0$;

(ii) $A_n \;\ge\; -\dfrac{\beta}{2}\, V_n - C_0$.

In particular $|A_n| \le \frac{\beta}{2} V_n + C_0$ for $n \ge n_0$.

Proof. (i) *(Lower bound on the evidence.)* Pointwise on M, with $t = nK_g(u)$, Cauchy–Schwarz gives $\beta\sqrt{t}\,\xi_n(u) \ge -\frac{\beta}{2} t - \frac{\beta}{2} \|\xi_n\|_\infty^2$, so

$$Z_n^0 \;\ge\; Z_n^{(1)} \;\ge\; e^{-\beta\|\xi_n\|_\infty^2/2} \int_M e^{-\frac{3}{2} n\beta K_g(u)}\, \omega(du).$$

The last integral equals $\frac{(\log n)^{m-1}}{(m-1)!\,n^\lambda}\,\Lambda_n[F]$ for the deterministic insertion $F(t,u)=e^{-3\beta t/2}$, which is admissible with $A=0$ and $N_0(F)=1$; by Theorem 5.2 and Proposition 3.2,

$$\Lambda_n[F]\;\longrightarrow\;\Lambda[F]\;=\;\Gamma(\lambda)\Big(\frac{2}{3\beta}\Big)^{\lambda}\mu(M)\;>\;0.$$

Choose n_0 so that $\Lambda_n[F]\ge c_0$ for $n\ge n_0$, with $c_0:=\frac12\Lambda[F]>0$; then $A_n\le\frac{\beta}{2}\|\xi_n\|_\infty^2+\log\big((m-1)!/c_0\big)$.

(ii) *(Upper bound on the evidence.)* Pointwise, $\beta\sqrt{t}\,\xi_n\le\frac{\beta}{2}t+\frac{\beta}{2}\|\xi_n\|_\infty^2$, so by Lemma 4.1 with the admissible $G(t)=e^{-\beta t/2}$,

$$Z_n^{(1)}\;\le\;e^{\beta\|\xi_n\|_\infty^2/2}\int_M e^{-\frac12 n\beta K_g}\,d\omega\;\le\;C\,e^{\beta\|\xi_n\|_\infty^2/2}\,\frac{(\log n)^{m-1}}{n^\lambda},$$

while on the outer region, as in the proof of Theorem 6.4, $Z_n^{(2)}\le e^{-n\beta\epsilon/2}\,e^{\frac{\beta}{2}\sup_{\{K\ge\epsilon\}}\psi_n^2}$. Enlarging n_0 so that $n^\lambda(\log n)^{1-m}e^{-n\beta\epsilon/2}\le1$, and using $Ce^p+e^q\le(C+1)\,e^{\max(p,q)}$,

$$\frac{n^\lambda}{(\log n)^{m-1}}\,Z_n^0\;\le\;C'\exp\Big(\frac{\beta}{2}\,V_n\Big),$$

and (ii) follows by taking $-\log$. □

Remark 7.5. Part (ii) is the coordinate-free counterpart of the estimate in the proof of [1, Main Thm. 6.2(2)]. Part (i) replaces the chart-level lower bounds [1, Thms. 4.8, 4.10]: in the present framework the lower bound on the evidence is not a separate computation but an instance of the kernel theorem for a deterministic insertion, combined with the positivity $\mu(M)>0$. The upper-bound-only character of §§4–5 is thereby sufficient after all.

7.4 The expectation version of the free energy formula

Theorem 7.6 (Expectation version; cf. [1, Main Thm. 6.2(2), Cor. 6.1(2)])**.** *Assume (I) with index $s=4$ and (II). Then $\{A_n\}_{n\ge n_0}$ is AUI with $\sup_{n\ge n_0}\mathbb{E}[|A_n|]<\infty$, the expectation $\mathbb{E}\big[|\log\mathcal{Y}(\xi)|\big]$ is finite, and*

$$\mathbb{E}\big[F_n^0\big]-\lambda\log n+(m-1)\log\log n\;\longrightarrow\;-\mathbb{E}\big[\log\mathcal{Y}(\xi)\big].$$

Consequently, if the entropy $S=-\int q\log q\,dx$ is finite,

$$\mathbb{E}[F_n]\;=\;n\beta S+\lambda\log n-(m-1)\log\log n-\mathbb{E}\big[\log\mathcal{Y}(\xi)\big]+o(1).$$

Proof. By Lemma 7.4, $|A_n|\le\frac{\beta}{2}V_n+C_0\le C_1(1+V_n)$ for $n\ge n_0$. By Lemma 7.3 and $V_n^2\le\|\xi_n\|_\infty^4+\sup_{\{K\ge\epsilon\}}\psi_n^4$, we have $\sup_n\mathbb{E}\big[(1+V_n)^2\big]<\infty$; hence $\{1+V_n\}$ is AUI by Theorem 7.2(b), and $\{A_n\}_{n\ge n_0}$ is AUI by Theorem 7.2(a), with $\mathbb{E}[|A_n|]\le C_1(1+\mathbb{E}[V_n])$ uniformly bounded. By Corollary 6.5, $A_n\to-\log\mathcal{Y}(\xi)$ in law; Theorem 7.2(c) then yields $\mathbb{E}[|\log\mathcal{Y}(\xi)|]<\infty$ and $\mathbb{E}[A_n]\to-\mathbb{E}[\log\mathcal{Y}(\xi)]$, which is the first display. The second follows from $F_n=n\beta S_n+F_n^0$ and $\mathbb{E}[S_n]=S$. □

Remark 7.7. The integrability of $\log\mathcal{Y}(\xi)$ can also be read off directly and two-sidedly: the Cauchy–Schwarz estimates used above give, for every $\eta\in C(M)$,

$$\frac{\mu(M)}{(m-1)!}\,e^{-\beta\|\eta\|_\infty^2/2}\int_0^\infty t^{\lambda-1}e^{-3\beta t/2}\,dt\;\le\;\mathcal{Y}(\eta)\;\le\;\frac{\mu(M)}{(m-1)!}\,e^{\beta\|\eta\|_\infty^2/2}\int_0^\infty t^{\lambda-1}e^{-\beta t/2}\,dt,$$

so $|\log\mathcal{Y}(\eta)|\le\frac{\beta}{2}\|\eta\|_\infty^2+C_{\mathcal{Y}}$; and $\|\xi\|_\infty^2$ has moments of every order by Fernique's theorem [7].

8 The four errors

Throughout this section condition (II) holds and condition (I) is assumed with index $s = 4$, strengthened to $s = 6$ for the Bayes pair and the statements derived from it (Lemma 8.22, Theorem 8.24, and §8.8). In [1], all of §6.3 is stated under the blanket assumption $s = 6$, so the $s = 4$ statements for the Gibbs pair below are slightly stronger than their counterparts there. Let $\langle \cdot \rangle$ denote the posterior expectation at inverse temperature β,

$$\langle H \rangle \ := \ \frac{\int_W H(w)\, e^{-n\beta K_n(w)}\, \varphi(w)\, dw}{\int_W e^{-n\beta K_n(w)}\, \varphi(w)\, dw},$$

and let X denote a fresh sample from q, independent of D_n.

8.1 The quartet and the main theorem

Definition 8.1 (cf. [1, Def. 6.5])**.** The *Bayes generalization error*, *Bayes training error*, *Gibbs generalization error* and *Gibbs training error* are

$$B_g := \mathbb{E}_X\big[-\log\langle e^{-f(X,w)}\rangle\big], \quad B_t := \frac{1}{n}\sum_{i=1}^{n}\big[-\log\langle e^{-f(X_i,w)}\rangle\big], \quad G_g := \langle K \rangle, \quad G_t := \langle K_n \rangle.$$

(Equivalently $B_g = \mathbb{E}_X[\log(q(X)/\hat{p}(X))]$ for the predictive density $\hat{p}(x) = \langle p(x \mid w)\rangle$, and G_g, G_t are the mean generalization and training errors of sampling w from the posterior.)

Theorem 8.2 (The four error formulas; cf. [1, Thm. 6.10, eqs. (6.27)–(6.30)])**.** *With $\nu = \nu(\beta)$ the singular fluctuation of Definition 8.9 (finite, and ≥ 0 by Remark 8.25),*

$$\mathbb{E}[G_g] = \frac{\lambda + \nu\beta}{\beta n} + o\Big(\frac{1}{n}\Big), \qquad \mathbb{E}[G_t] = \frac{\lambda - \nu\beta}{\beta n} + o\Big(\frac{1}{n}\Big)$$

under index $s = 4$, and

$$\mathbb{E}[B_g] = \frac{\lambda + \nu\beta - \nu}{\beta n} + o\Big(\frac{1}{n}\Big), \qquad \mathbb{E}[B_t] = \frac{\lambda - \nu\beta - \nu}{\beta n} + o\Big(\frac{1}{n}\Big)$$

under index $s = 6$.

Proof. The Gibbs pair is Theorem 8.15; the Bayes pair is Theorem 8.24. □

Corollary 8.3 (Equations of states in mean; cf. [1, Main Thm. 6.3])**.** *Under index $s = 6$,*

$$\mathbb{E}[B_g] - \mathbb{E}[B_t] \ = \ 2\beta\big(\mathbb{E}[G_t] - \mathbb{E}[B_t]\big) + o\Big(\frac{1}{n}\Big), \qquad \mathbb{E}[G_g] - \mathbb{E}[G_t] \ = \ 2\beta\big(\mathbb{E}[G_t] - \mathbb{E}[B_t]\big) + o\Big(\frac{1}{n}\Big).$$

Proof. By Theorem 8.2, $\mathbb{E}[B_g] - \mathbb{E}[B_t] = \frac{2\nu}{n} + o(\frac{1}{n})$, $\mathbb{E}[G_g] - \mathbb{E}[G_t] = \frac{2\nu}{n} + o(\frac{1}{n})$, and $\mathbb{E}[G_t] - \mathbb{E}[B_t] = \frac{\nu}{\beta n} + o(\frac{1}{n})$. □

8.2 The renormalized posterior

For $A \geq 0$ let $\mathcal{P}_A$ be the class of jointly measurable $F : [0,\infty) \times M \to \mathbb{R}$ with $F(t,\cdot) \in C^1(M)$ and

$$P_A(F) \;:=\; \sup_{t,u}\,(1+t)^{-A}\big(|F(t,u)| + |\nabla_u F(t,u)|\big) \;<\; \infty$$

(polynomial growth; no decay in t is required). Write $E_\eta(t,u) := e^{-\beta t+\beta\sqrt{t}\,\eta(u)}$ and define the *renormalized posterior average* and its limit,

$$\langle F\rangle_n \;:=\; \frac{\Lambda_n[F\,E_{\xi_n}]}{\Lambda_n[E_{\xi_n}]}, \qquad \langle\!\langle F\rangle\!\rangle_\eta \;:=\; \frac{\Lambda[F\,E_\eta]}{\Lambda[E_\eta]} \;=\; \frac{1}{(m-1)!\,\mathcal{Y}(\eta)}\int_0^\infty\!\!\int_\Sigma F(t,u)\,t^{\lambda-1}E_\eta(t,u)\,d\mu\,dt.$$

By (3) and the standard form (5), $\langle F\rangle_n$ is exactly the posterior average, restricted and renormalized to W_ϵ, of $w = g(u) \mapsto F(nK_g(u),u)$; and $\langle\!\langle\cdot\rangle\!\rangle_\eta$ is the expectation under the probability kernel $t^{\lambda-1}E_\eta\,d\mu\,dt$ normalized by $(m-1)!\,\mathcal{Y}(\eta) > 0$.

Proposition 8.4 (Posterior kernel theorem)**.** *Set $\Theta_n := (1+\|\xi_n\|_{C^1})^2\,e^{2\beta\|\xi_n\|_\infty^2}$ and let r_n' denote the rate of Theorem 5.2. There are constants C_*, C such that:*

(i) on the event $\Omega_n := \{C_*\Theta_n r_n' \leq \frac{1}{2}\}$, whose probability tends to 1,

$$\big|\langle F\rangle_n - \langle\!\langle F\rangle\!\rangle_{\xi_n}\big| \;\leq\; C\,P_A(F)\,\Theta_n\,r_n' \qquad \textit{for every } F \in \mathcal{P}_A;$$

(ii) for fixed $F \in \mathcal{P}_A$, the map $\eta \mapsto \langle\!\langle F\rangle\!\rangle_\eta$ is continuous on $C(M)$; consequently, for $F_1,\dots,F_k \in \mathcal{P}_A$, $(\langle F_j\rangle_n)_j \to (\langle\!\langle F_j\rangle\!\rangle_\xi)_j$ jointly in law;

(iii) the same statements hold for the two-replica averages on $M\times M$: with $\langle H\rangle_n^{(2)} := \Lambda_n \otimes \Lambda_n[H\,E_{\xi_n}\otimes E_{\xi_n}]/\Lambda_n[E_{\xi_n}]^2$ and H jointly measurable, C^1 in (u,v), of polynomial growth in (t,t'), one has $\langle H\rangle_n^{(2)} - \langle\!\langle H\rangle\!\rangle_{\xi_n}^{(2)} \to 0$ in probability, where $\langle\!\langle\cdot\rangle\!\rangle_\eta^{(2)}$ is the expectation under the product kernel.

Proof. From $\beta\sqrt{t}\,\eta \leq \frac{\beta}{2}t + \frac{\beta}{2}\eta^2$ and $\nabla_u(FE_\eta) = (\nabla_u F)E_\eta + \beta\sqrt{t}\,(\nabla_u\eta)FE_\eta$,

$$N_{A+1/2}\big(F\,E_{\xi_n}\big) \;\leq\; C\,P_A(F)\,\big(1+\beta\|\xi_n\|_{C^1}\big)\,e^{\beta\|\xi_n\|_\infty^2/2}. \tag{13}$$

Theorem 5.2 then bounds $|\Lambda_n[FE_{\xi_n}] - \Lambda[FE_{\xi_n}]|$ and $|\Lambda_n[E_{\xi_n}] - \Lambda[E_{\xi_n}]|$ by $CP_A(F)(1+\|\xi_n\|_{C^1})e^{\beta\|\xi_n\|^2/2}\,r_n'$. The denominator limit satisfies, by the display of Remark 7.7,

$$\Lambda[E_{\xi_n}] \;=\; (m-1)!\,\mathcal{Y}(\xi_n) \;\geq\; c_1\,e^{-\beta\|\xi_n\|_\infty^2/2}, \qquad \big|\langle\!\langle F\rangle\!\rangle_{\xi_n}\big| \;\leq\; C\,P_A(F)\,e^{2\beta\|\xi_n\|_\infty^2},$$

the second bound by the same Cauchy–Schwarz estimates. On Ω_n (with C_* chosen from these constants), $\Lambda_n[E_{\xi_n}] \geq \frac{1}{2}\Lambda[E_{\xi_n}]$, and the elementary ratio inequality $|\frac{a}{b} - \frac{a'}{b'}| \leq \frac{|a-a'|}{b} + |\frac{a'}{b'}|\frac{|b-b'|}{b}$ gives (i); $\Pr(\Omega_n) \to 1$ since Θ_n is bounded in probability (Theorem 2.4, Lemma 6.1) while $r_n' \to 0$. For (ii), continuity follows by dominated convergence (majorant $(1+t)^A t^{\lambda-1}e^{-\beta t/2}e^{\beta R^2/2}$ on $\{\|\eta\| \leq R\}$) and $\mathcal{Y} > 0$; the joint convergence in law then follows from (i), Slutsky, and the continuous mapping theorem. For (iii), apply Theorem 5.2 twice via Fubini — first in (t,u) for fixed (t',v), with norms uniform in (t',v) by hypothesis, then in (t',v) — to get the two-replica kernel convergence with the same structure of bounds; the denominator is the square of the single-replica one. □

8.3 Tilted free energies and their derivatives

Under condition (I), K is continuous and bounded on W (it is the $L^2(q)$-pairing of an analytic family with q, and $|K| \le \mathbb{E}_X[M]$); set $K_{\max} := \max_W K$. For (γ, s) near $(1,0)$ define the *tilted partition function*

$$Z_n(\gamma, s) \ := \ \int_W e^{-n\beta(\gamma K_n(w)+sK(w))}\,\varphi(w)\,dw,$$

so $Z_n(1,0) = Z_n^0$, and note that $\gamma \mapsto \log Z_n(\gamma,s)$ and $s \mapsto \log Z_n(\gamma,s)$ are convex (log-partition functions in their couplings), with

$$-\partial_\gamma \log Z_n(\gamma,0)\big|_{\gamma=1} = \langle n\beta K_n\rangle = n\beta\, G_t, \qquad -\partial_s \log Z_n(1,s)\big|_{s=0} = \langle n\beta K\rangle = n\beta\, G_g. \quad (14)$$

Lemma 8.5 (Tilted free energy asymptotics)**.** *There is an open neighborhood $U \ni (1,0)$ such that for every $(\gamma,s) \in U$,*

$$\Phi_n(\gamma,s) \ := \ \mathbb{E}\log Z_n(\gamma,s) + \lambda\log n - (m-1)\log\log n \ \longrightarrow \ \Phi_\infty(\gamma,s) \ := \ \mathbb{E}\log Y_{\gamma,s}(\xi),$$

where $Y_{\gamma,s}(\xi) := \frac{1}{(m-1)!}\int_0^\infty\int_\Sigma t^{\lambda-1}e^{-(\gamma+s)\beta t+\gamma\beta\sqrt{t}\,\xi(u)}\,d\mu\,dt$. The same holds for the inner partition function $Z_n^{(1)}(\gamma,s)$ (integral over W_ϵ).

Proof. This is the proof of Theorems 6.4 and 7.6 run with the insertion $F_n = e^{-(\gamma+s)\beta t+\gamma\beta\sqrt{t}\,\xi_n(u)}$. Two points need checking. First, admissibility: by Cauchy–Schwarz the exponent is at most $-(\frac{\gamma}{2}+s)\beta t + \frac{\gamma\beta}{2}\|\xi_n\|^2$, so on a small enough U the insertions are admissible for the weight $(1+t)^A e^{-\tilde\beta t/2}$ with a fixed $\tilde\beta > 0$; Lemmas 4.1–4.2 and Theorem 5.2 hold verbatim for any fixed weight parameter $\tilde\beta > 0$ (nothing in their proofs used the specific value), with constants depending on $\tilde\beta$. Both directions of the envelope (Lemma 7.4) go through with the same substitutions. Second, the outer region: $e^{-sn\beta K} \le e^{|s|n\beta K_{\max}}$ there, and by (6), $\gamma n\beta K_n \ge \frac{\gamma\beta}{2}(n\epsilon - \sup\psi_n^2)$, so the outer contribution carries the factor $e^{-n\beta(\gamma\epsilon/2-|s|K_{\max})}$ and remains exponentially negligible for $|s| < \gamma\epsilon/(2K_{\max})$; shrink U accordingly. For $Z_n^{(1)}$ there is no outer region and the argument is shorter. ☐

Lemma 8.6 (Derivatives of the limit)**.** *The partial functions $\gamma \mapsto \Phi_\infty(\gamma,0)$ and $s \mapsto \Phi_\infty(1,s)$ are differentiable at $\gamma = 1$, resp. $s = 0$, with*

$$\partial_\gamma\Phi_\infty(1,0) \ = \ -\beta\,\mathbb{E}\big[\langle\!\langle t - \sqrt{t}\,\xi(u)\rangle\!\rangle_\xi\big], \qquad \partial_s\Phi_\infty(1,0) \ = \ -\beta\,\mathbb{E}\big[\langle\!\langle t\rangle\!\rangle_\xi\big],$$

and in particular both expectations are finite.

Proof. Pathwise, $\gamma \mapsto \log Y_{\gamma,0}(\xi)$ is a smooth, convex log-partition function in the coupling γ of the energy $\beta(t - \sqrt{t}\,\xi(u))$, with $\partial_\gamma \log Y_{\gamma,0}|_{\gamma=1} = -\beta\langle\!\langle t - \sqrt{t}\,\xi\rangle\!\rangle_\xi$. By convexity the difference quotients $h^{-1}(\log Y_{1+h,0} - \log Y_{1,0})$ are nondecreasing in h and converge pointwise, as $h \downarrow 0$, to the pathwise derivative; they are integrable at any fixed h_0 (the two-sided bounds of Remark 7.7 hold at each parameter in U, and $\mathbb{E}\|\xi\|_\infty^2 < \infty$). By monotone convergence the right derivative of $\Phi_\infty(\cdot,0)$ at 1 equals $\mathbb{E}[\partial_\gamma \log Y_{\gamma,0}|_1]$, with value in $[-\infty,\infty)$; the same argument from the left gives the left derivative, with value in $(-\infty,\infty]$, equal to the same expectation. Since $\Phi_\infty(\cdot,0)$ is finite and convex on a neighborhood of 1, its one-sided derivatives at 1 are finite; hence they coincide, are finite, and equal $-\beta\mathbb{E}\langle\!\langle t - \sqrt{t}\,\xi\rangle\!\rangle_\xi$. The argument in s is identical with the energy βt. ☐

8.4 The singular fluctuation

Lemma 8.7 (Integration by parts in t)**.** *Surely, for every $\eta \in C(M)$,*

$$\beta\,\langle\!\langle t\rangle\!\rangle_\eta \;=\; \lambda + \frac{\beta}{2}\,\langle\!\langle \sqrt{t}\,\eta(u)\rangle\!\rangle_\eta. \tag{15}$$

Proof. For fixed u, integrating $\frac{d}{dt}\big[t^\lambda e^{-\beta t+\beta\eta(u)\sqrt{t}}\big]$ over $(0,\infty)$ (boundary terms vanish since $\lambda>0$) gives $0=\lambda\int t^{\lambda-1}E_\eta\,dt-\beta\int t^\lambda E_\eta\,dt+\frac{\beta\eta(u)}{2}\int t^{\lambda-\frac12}E_\eta\,dt$; mix over u with $d\mu$ and divide by $(m-1)!\,\mathcal{Y}(\eta)$. ☐

Corollary 8.8 (Sure moment envelopes)**.** *Surely, for every $\eta \in C(M)$ (equivalently every $\eta\in C(\Sigma)$, the kernel being carried by Σ),*

$$\langle\!\langle t\rangle\!\rangle_\eta \;\le\; \frac{2\lambda}{\beta}+\frac{\|\eta\|_\infty^2}{2},\qquad \langle\!\langle \sqrt{t}\rangle\!\rangle_\eta \;\le\; \langle\!\langle t\rangle\!\rangle_\eta^{1/2},\qquad \big|\langle\!\langle \sqrt{t}\,\eta(u)\rangle\!\rangle_\eta\big| \;\le\; \|\eta\|_\infty\,\langle\!\langle t\rangle\!\rangle_\eta^{1/2}.$$

Proof. The second and third bounds are Jensen's inequality. For the first, Lemma 8.7 and the third bound give $\beta\langle\!\langle t\rangle\!\rangle_\eta \le \lambda+\frac{\beta}{2}\|\eta\|_\infty\langle\!\langle t\rangle\!\rangle_\eta^{1/2}$; solving the quadratic inequality in $\langle\!\langle t\rangle\!\rangle_\eta^{1/2}$ yields $\langle\!\langle t\rangle\!\rangle_\eta^{1/2}\le\frac12\|\eta\|_\infty+\sqrt{\lambda/\beta}$. ☐

Definition 8.9. The *singular fluctuation* at inverse temperature β is $\nu(\beta):=\frac12\,\mathbb{E}\big[\langle\!\langle \sqrt{t}\,\xi(u)\rangle\!\rangle_\xi\big]$.

Proposition 8.10. *$\nu(\beta)$ is finite, and*

$$\mathbb{E}\big[\langle\!\langle t\rangle\!\rangle_\xi\big]=\frac{\lambda}{\beta}+\nu(\beta),\qquad \mathbb{E}\big[\langle\!\langle t-\sqrt{t}\,\xi(u)\rangle\!\rangle_\xi\big]=\frac{\lambda}{\beta}-\nu(\beta).$$

Moreover, under the dictionary $S_\lambda(c)=\int_0^\infty t^{\lambda-1}e^{-\beta t+\beta c\sqrt{t}}dt$, $S'_\lambda=\beta S_{\lambda+1/2}$, Definition 8.9 coincides with [1, Def. 5.9] applied to the compact set Σ, the measure μ, and the process $\xi|_\Sigma$; the standing hypothesis there, $\mathbb{E}[\xi(u)^2]=2$ on Σ, holds by Theorem 2.4 since $\Sigma\subseteq g^{-1}(W_0)$.

Proof. By Lemma 8.6, $\mathbb{E}\langle\!\langle t\rangle\!\rangle$ and $\mathbb{E}\langle\!\langle t-\sqrt{t}\xi\rangle\!\rangle$ are finite. Identity (15) at $\eta=\xi$ reads $\langle\!\langle \sqrt{t}\,\xi\rangle\!\rangle_\xi = 2\langle\!\langle t\rangle\!\rangle_\xi-2\lambda/\beta$ surely; the right side is integrable, so ν is finite and the two displayed identities follow by taking expectations. (Alternatively, and under index $s=2$ only: Corollary 8.8 bounds $\langle\!\langle t\rangle\!\rangle_\xi$ and $|\langle\!\langle \sqrt{t}\,\xi\rangle\!\rangle_\xi|$ surely by $C(1+\|\xi\|_\infty^2)$, integrable by Fernique [7], so the same conclusions follow without Lemma 8.6; this is the reading used in Remark 8.14.) The dictionary is a direct rewriting: $\int_\Sigma \mu(du)\,\xi(u)S'_\lambda(\xi(u))/\int_\Sigma\mu(du)S_\lambda(\xi(u))=\beta\,\langle\!\langle \sqrt{t}\,\xi(u)\rangle\!\rangle_\xi$. ☐

Remark 8.11. Since $\partial_\beta\log\big[(m-1)!\,\mathcal{Y}_\beta(\xi)\big]=\langle\!\langle -t+\sqrt{t}\,\xi\rangle\!\rangle_\xi$ pathwise (with $\mathcal{Y}_\beta$ making the β-dependence explicit), Proposition 8.10 states that the mean limiting free energy $F(\beta):=-\mathbb{E}\log\big[(m-1)!\mathcal{Y}_\beta(\xi)\big]$ satisfies $\partial F/\partial\beta=\lambda/\beta-\nu(\beta)$, recovering [1, Rem. 5.9(1)].

Remark 8.12 (Bound on the singular fluctuation; cf. [1, Thm. 6.10]). Write $\kappa:=\mathbb{E}\big[\|\xi\|_\infty^2\big]$, finite by Fernique's theorem [7]. Cauchy–Schwarz under the kernel $\langle\!\langle\cdot\rangle\!\rangle_\xi$ and then under $\mathbb{E}$ gives, with Proposition 8.10,

$$\nu \;=\; \tfrac12\,\mathbb{E}\big[\langle\!\langle \sqrt{t}\,\xi(u)\rangle\!\rangle_\xi\big] \;\le\; \tfrac12\,\mathbb{E}\big[\langle\!\langle t\rangle\!\rangle_\xi^{1/2}\,\|\xi\|_\infty\big] \;\le\; \tfrac12\,\big(\mathbb{E}\langle\!\langle t\rangle\!\rangle_\xi\big)^{1/2}\kappa^{1/2} \;=\; \tfrac12\,\big(\lambda/\beta+\nu\big)^{1/2}\,\kappa^{1/2}.$$

Solving the quadratic inequality $\nu^2\le\frac{\kappa}{4}(\lambda/\beta+\nu)$,

$$\nu(\beta)\;\le\;\frac{\kappa}{8}+\frac{\kappa^{1/2}}{8}\sqrt{\kappa+16\lambda/\beta}\,,$$

which is the bound of [1, Thm. 6.10]. (Nonnegativity of ν is Remark 8.25 below.)

Theorem 8.13 (Gaussian integration by parts; = [1, Thm. 5.11, eq. (5.24)] on $(\Sigma, \mu, \xi|_\Sigma)$)**.**

$$\frac{\beta^2}{2}\,\mathbb{E}\,\mathbb{E}_X\Big[\langle\!\langle \sqrt{t}\,a(X,u)\rangle\!\rangle_\xi^2\Big] \;=\; \lambda\beta + (\beta^2-\beta)\,\nu(\beta).$$

Proof. All kernels are carried by Σ, so every functional $\langle\!\langle\cdot\rangle\!\rangle_\eta$ depends on η only through $\eta|_\Sigma$, and we work on $C(\Sigma)$ throughout. The proof uses only the following structure: μ is a finite positive measure on the compact set Σ; $u \mapsto a_u := a(\cdot,u)$ is continuous from Σ to $L^2(q)$ with $|a(x,u)| \le c\,M(x)$, $M \in L^2(q)$; and $\xi|_\Sigma$ is a centred Gaussian process with almost surely continuous paths and covariance

$$\mathbb{E}[\xi(u)\xi(v)] \;=\; \rho(u,v) \;=\; \langle a_u, a_v\rangle_{L^2(q)}, \qquad \rho(u,u) = 2 \;\text{ on } \Sigma$$

(Theorem 2.4).

Step 1: series representation. Let $H := \overline{\operatorname{span}}\{a_u : u \in \Sigma\} \subseteq L^2(q)$, a separable Hilbert space, and $\{e_j\}_{j\ge1}$ a complete orthonormal system of H (finite if $\dim H < \infty$, in which case the limits below are trivial). The map $\sum_i c_i a_{u_i} \mapsto \sum_i c_i \xi(u_i)$ preserves inner products, hence extends to a linear isometry $J : H \to L^2(\mathrm{Pr})$ whose image consists of centred, jointly Gaussian variables; the $g_j := J(e_j)$ are orthonormal and jointly Gaussian, hence i.i.d. standard normal. Set

$$\alpha_j(u) := \langle a_u, e_j\rangle \in C(\Sigma), \qquad \xi_N := \sum_{j\le N} g_j\,\alpha_j, \qquad \rho_N(u,v) := \sum_{j\le N} \alpha_j(u)\alpha_j(v) = \langle P_N a_u, a_v\rangle,$$

with P_N the projection onto $\operatorname{span}(e_1,\dots,e_N)$; note $\|\alpha_j\|_\infty \le \sup_u \|a_u\| = \sqrt{2}$. Since $\mathbb{E}[\xi(u)g_j] = \langle a_u, e_j\rangle = \alpha_j(u)$ and all variables are jointly Gaussian, $\xi_N(u) = \mathbb{E}[\xi(u) \mid \mathcal{F}_N]$ for $\mathcal{F}_N := \sigma(g_1,\dots,g_N)$, so $(\xi_N)_N$ is a $C(\Sigma)$-valued martingale closed by ξ, which is measurable with respect to (the completion of) $\mathcal{F}_\infty$ because $\xi(u) = \lim_N J(P_N a_u)$ in $L^2(\mathrm{Pr})$ and the paths are continuous. By the martingale convergence theorem in separable Banach spaces (e.g. [8, Ch. 2]),

$$\|\xi_N - \xi\|_\infty \;\longrightarrow\; 0 \quad \text{almost surely}, \qquad \|\xi_N\|_\infty \;\le\; \mathbb{E}\big[\|\xi\|_\infty \mid \mathcal{F}_N\big],$$

the second by the conditional Jensen inequality; in particular $\sup_N \mathbb{E}\|\xi_N\|_\infty^4 \le \mathbb{E}\|\xi\|_\infty^4 < \infty$ by Fernique's theorem [7]. Moreover $\{a_u : u \in \Sigma\}$ is compact in H, so $\sup_u \|(I-P_N)a_u\| \to 0$, whence $\rho_N \to \rho$ uniformly on $\Sigma\times\Sigma$; and $|\rho_N(u,v)| \le \rho_N(u,u)^{1/2}\rho_N(v,v)^{1/2} \le 2$.

Step 2: the finite-dimensional identity. Let $F_N(g_1,\dots,g_N) := \log\big[(m-1)!\,\mathcal{Y}(\xi_N)\big]$, a smooth function of $(g_1,\dots,g_N)$: differentiation under the integral sign is justified by dominated convergence, each g-derivative multiplying the kernel by a factor $\beta\sqrt{t}\,\alpha_j(u)$ dominated, for $\|\xi_N\|_\infty \le R$, by $C\sqrt{t}\,e^{\beta t/2+\beta R^2/2}$ against $t^{\lambda-1}e^{-\beta t}$. Explicitly,

$$\partial_j F_N = \beta\,\langle\!\langle \sqrt{t}\,\alpha_j(u)\rangle\!\rangle_{\xi_N}, \qquad \partial_j^2 F_N = \beta^2\Big(\langle\!\langle t\,\alpha_j(u)^2\rangle\!\rangle_{\xi_N} - \langle\!\langle \sqrt{t}\,\alpha_j(u)\rangle\!\rangle_{\xi_N}^2\Big).$$

By Corollary 8.8 and $\|\xi_N\|_\infty \le \sqrt{2}\sum_{j\le N}|g_j|$, both are of polynomial growth in $(g_1,\dots,g_N)$. For a standard Gaussian vector and a C^1 function of polynomial growth, $\mathbb{E}[g_jF(g)] = \mathbb{E}[\partial_jF(g)]$ (one-dimensional integration by parts in the j-th coordinate, the boundary terms vanishing); applying this with $F = \partial_j F_N$ and summing over $j \le N$, using $\sum_{j\le N} g_j\alpha_j = \xi_N$ on the left and $\sum_{j\le N}\langle\!\langle\sqrt{t}\,\alpha_j(u)\rangle\!\rangle\langle\!\langle\sqrt{t'}\,\alpha_j(v)\rangle\!\rangle = \langle\!\langle\sqrt{t}\sqrt{t'}\,\rho_N(u,v)\rangle\!\rangle^{(2)}$ on the right,

$$\mathbb{E}\Big[\beta\,\langle\!\langle\sqrt{t}\,\xi_N(u)\rangle\!\rangle_{\xi_N}\Big] \;=\; \beta^2\,\mathbb{E}\Big[\langle\!\langle t\,\rho_N(u,u)\rangle\!\rangle_{\xi_N}\Big] - \beta^2\,\mathbb{E}\Big[\langle\!\langle\sqrt{t}\sqrt{t'}\,\rho_N(u,v)\rangle\!\rangle_{\xi_N}^{(2)}\Big].$$

Step 3: passage to the limit. By Corollary 8.8, $|\rho_N| \le 2$ and Jensen, the three integrands are bounded, surely, by $C(1 + \|\xi_N\|_\infty^2)$, hence bounded in $L^2(\mathrm{Pr})$ uniformly in N by Step 1, hence uniformly integrable. Almost surely $\xi_N \to \xi$ and $\rho_N \to \rho$ uniformly, and the integrands converge: replacing ρ_N by ρ costs at most $\|\rho_N - \rho\|_\infty \langle\!\langle t \rangle\!\rangle_{\xi_N} \to 0$, while for fixed continuous ϱ the maps $\eta \mapsto \langle\!\langle \sqrt{t}\,\eta(u) \rangle\!\rangle_\eta$, $\langle\!\langle t\,\varrho(u,u) \rangle\!\rangle_\eta$, $\langle\!\langle \sqrt{t}\sqrt{t'}\varrho(u,v) \rangle\!\rangle_\eta^{(2)}$ are continuous for the uniform norm on bounded sets, by dominated convergence and the lower bound on $\mathcal{Y}$ of Remark 7.7 (as in Proposition 8.4(ii)). By Vitali's theorem the identity passes to the limit; using $\rho(u,u) = 2$ on Σ,

$$\mathbb{E}\Big[\beta\,\langle\!\langle \sqrt{t}\,\xi(u) \rangle\!\rangle_\xi\Big] \;=\; 2\beta^2\,\mathbb{E}\big[\langle\!\langle t \rangle\!\rangle_\xi\big] - \beta^2\,\mathbb{E}\Big[\langle\!\langle \sqrt{t}\sqrt{t'}\,\rho(u,v) \rangle\!\rangle_\xi^{(2)}\Big].$$

The left side is $2\beta\nu$ by Definition 8.9; $\mathbb{E}\langle\!\langle t \rangle\!\rangle_\xi = \lambda/\beta + \nu$ by Proposition 8.10; and by Tonelli (the integrand is dominated by $c^2 M(X)^2 \langle\!\langle t \rangle\!\rangle_\xi$, jointly integrable),

$$\mathbb{E}\Big[\langle\!\langle \sqrt{t}\sqrt{t'}\,\rho(u,v) \rangle\!\rangle_\xi^{(2)}\Big] \;=\; \mathbb{E}\,\mathbb{E}_X\Big[\langle\!\langle \sqrt{t}\,a(X,u) \rangle\!\rangle_\xi^2\Big].$$

Rearranging $2\beta\nu = 2\beta^2(\lambda/\beta + \nu) - \beta^2\,\mathbb{E}\mathbb{E}_X\big[\langle\!\langle \sqrt{t}a \rangle\!\rangle^2\big]$ gives the statement. □

Remark 8.14. Theorem 8.13 concerns the limit objects only, so it holds under the index $s = 2$ conditions, and its proof uses only the structure listed at its start together with Corollary 8.8 and Proposition 8.10 in its envelope-based reading — not the kernel theorem, and nothing chart-related. The hypothesis $\rho(u,u) = 2$ enters only through $\langle\!\langle t\,\rho(u,u) \rangle\!\rangle_\xi = 2\langle\!\langle t \rangle\!\rangle_\xi$; without it the identity reads $2\beta\nu = \beta^2\,\mathbb{E}\langle\!\langle t\,\rho(u,u) \rangle\!\rangle_\xi - \beta^2\,\mathbb{E}\mathbb{E}_X\langle\!\langle \sqrt{t}a \rangle\!\rangle_\xi^2$. The same scheme — represent the field, apply the finite-dimensional Stein identity to smooth functionals with polynomially enveloped derivatives, pass to the limit with the sure envelopes — computes further Gaussian functionals of the quartet limits, for instance the covariances needed for a second-order analysis of the estimators of §8.8; each such identity is one more application of the same calculus, not a new import.

8.5 The Gibbs pair

Theorem 8.15 (Gibbs errors)**.** *Under index $s = 4$,*

$$n\,\mathbb{E}[G_t] \;\longrightarrow\; \frac{\lambda}{\beta} - \nu(\beta), \qquad n\,\mathbb{E}[G_g] \;\longrightarrow\; \frac{\lambda}{\beta} + \nu(\beta).$$

Proof. The functions $\gamma \mapsto \Phi_n(\gamma, 0)$ are finite ($|\log Z_n(\gamma,0)| \le \gamma\beta\sum_i M(X_i)$, integrable), convex (expectations of pathwise convex functions), and differentiable, with $\Phi_n'(1) = -\mathbb{E}\langle n\beta K_n \rangle$; the differentiation under $\mathbb{E}$ is dominated by $\sup_\gamma |\partial_\gamma \log Z_n(\gamma,0)| \le \beta\sum_i M(X_i)$. By Lemma 8.5 they converge pointwise on a neighborhood of $\gamma = 1$ to the convex function $\Phi_\infty(\cdot, 0)$, which is differentiable at 1 by Lemma 8.6. Pointwise convergence of finite convex functions implies convergence of derivatives at every differentiability point of the limit (sandwich the derivatives between difference quotients on either side; see e.g. [9, Thm. 25.7]). Hence, using (14) and Proposition 8.10,

$$n\beta\,\mathbb{E}[G_t] = -\Phi_n'(1) \;\longrightarrow\; -\partial_\gamma\Phi_\infty(1,0) = \beta\,\mathbb{E}\langle\!\langle t - \sqrt{t}\,\xi \rangle\!\rangle_\xi = \lambda - \beta\nu.$$

The argument for G_g is identical in the variable s, with domination $|\partial_s \log Z_n(1,s)| \le n\beta K_{\max}$ (a constant for each n), giving $n\beta\,\mathbb{E}[G_g] \to -\partial_s\Phi_\infty(1,0) = \beta\mathbb{E}\langle\!\langle t \rangle\!\rangle_\xi = \lambda + \beta\nu$. □

Remark 8.16. The Gibbs pair required no posterior asymptotics: only the free energy expansion of Sections 6–7 at tilted couplings, plus convexity. The mechanism is that (λ, m) do not depend on the couplings, so the $\lambda \log n$ terms cancel in the derivatives. This appears to be a genuine simplification over the route through posterior moments.

The same convexity trick yields a moment bound needed for the Bayes pair.

Lemma 8.17 (Energy moments)**.** *Under index* $s = 4$, $\sup_n \mathbb{E}\big[\langle t\rangle_n^2\big] < \infty$ *and* $\sup_n \mathbb{E}\big[\langle t - \sqrt{t}\,\xi_n\rangle_n^2\big] < \infty$.

Proof. Let $\ell_n(\gamma) := \log Z_n^{(1)}(\gamma, 0)$, a convex function of γ with $\ell_n'(1) = -\beta\langle t - \sqrt{t}\,\xi_n\rangle_n$ (by (5), $n\beta K_n \circ g = \beta(t - \sqrt{t}\,\xi_n)$). Sandwiching the derivative between difference quotients at $h = \frac{1}{2}$, and noting that in the differences $\ell_n(\gamma) - \ell_n(\gamma')$ the terms $\lambda \log n - (m-1)\log\log n$ cancel (they are the same at every coupling),

$$\beta\,\big|\langle t - \sqrt{t}\,\xi_n\rangle_n\big| \;\le\; 2\sum_{\gamma\in\{1/2,\,1,\,3/2\}} \big|A_n^{(1),\gamma}\big|, \qquad A_n^{(1),\gamma} := -\log\Big(\tfrac{n^\lambda}{(\log n)^{m-1}} Z_n^{(1)}(\gamma, 0)\Big).$$

The envelope Lemma 7.4 applies to $Z_n^{(1)}(\gamma, 0)$ at each of the three couplings (part (i) verbatim at inverse temperature $\gamma\beta$; part (ii) with no outer region), giving $|A_n^{(1),\gamma}| \le C(1 + \|\xi_n\|_\infty^2)$ surely for $n \ge n_0$; hence $\mathbb{E}[\langle t - \sqrt{t}\xi_n\rangle_n^2] \le C(1 + \mathbb{E}\|\xi_n\|_\infty^4) \le C'$ by Lemma 7.3. Finally $t \le 2(t - \sqrt{t}\,\xi_n(u)) + \xi_n(u)^2$ pointwise, so $\langle t\rangle_n \le 2\langle t - \sqrt{t}\xi_n\rangle_n + \|\xi_n\|_\infty^2$. (For the finitely many $n < n_0$, $\langle t\rangle_n \le n\epsilon$ deterministically.) □

8.6 Sure moment envelopes at finite n

Lemma 8.17 controlled the first renormalized moment. We now control all of them, with sure polynomial envelopes matching those of Corollary 8.8 at the limit kernel. The mechanism is not an integration by parts at finite n: it combines the nonnegativity of t with an exponential insertion of optimized, vanishing rate, whose logarithm is a short coupling-difference of free energies and therefore inherits the first-derivative envelopes.

Theorem 8.18 (Sure moment envelopes at finite n)**.** *There are deterministic n_0 and $C_\star$, depending only on β and the resolution, such that, with $\mathcal{E}_n := C_\star\big(1 + \|\xi_n\|_\infty^2\big)$, surely for all $n \ge n_0$:*

(i) (exponential insertions of vanishing rate) for every $\delta \in (0, \frac{1}{8}]$,

$$\big\langle e^{\delta\beta t}\big\rangle_n \;\le\; e^{\delta\beta\,\mathcal{E}_n};$$

(ii) for every real $r > 0$,

$$\big\langle t^r\big\rangle_n \;\le\; \big(\mathcal{E}_n + 8r/\beta\big)^r \;\le\; \Big(\big(C_\star + \tfrac{8r}{\beta}\big)\big(1 + \|\xi_n\|_\infty^2\big)\Big)^r.$$

Proof. Step 1: a first-moment envelope, uniform in the coupling. For $s \in [-\frac{1}{4}, \frac{1}{4}]$ let $Z^{(1)}(1, s)$ be the inner tilted partition function of §8.3, $\ell(s) := \log Z^{(1)}(1, s)$, and let $\langle\cdot\rangle_{n,s}$ denote the renormalized posterior with weight $e^{-(1+s)\beta t + \beta\sqrt{t}\,\xi_n(u)}$, so that $\langle\cdot\rangle_{n,0} = \langle\cdot\rangle_n$, ℓ is convex and smooth, and $\ell'(s) = -\beta\langle t\rangle_{n,s}$ (differentiation under the integral is immediate, $t \le n\epsilon$ being bounded). Cauchy–Schwarz gives, pointwise in s,

$$e^{-\beta\|\xi_n\|_\infty^2/2}\int_M e^{-(\frac{3}{2}+s)\beta n K_g}\,d\omega \;\le\; Z^{(1)}(1,s) \;\le\; e^{\beta\|\xi_n\|_\infty^2/2}\int_M e^{-(\frac{1}{2}+s)\beta n K_g}\,d\omega.$$

On the range considered, $\frac{3}{2}+s \in [\frac{5}{4},\frac{7}{4}]$ and $\frac{1}{2}+s \in [\frac{1}{4},\frac{3}{4}]$. The deterministic insertions $e^{-(\frac{3}{2}+s)\beta t}$ form a compact family with N_0-norm at most 1 for the weight $e^{-\beta t/2}$ and limits $\Lambda\big[e^{-(\frac{3}{2}+s)\beta t}\big] = \Gamma(\lambda)\big((\frac{3}{2}+s)\beta\big)^{-\lambda}\mu(M)$ bounded below; Theorem 5.2, whose error is uniform over the family, therefore bounds the left integral below by $c\,(\log n)^{m-1}n^{-\lambda}$ for all such s and $n \ge n_0$, with deterministic $c>0$, n_0. Lemma 4.1 with weight parameter $\beta/2$ bounds the right integral above by $C\,(\log n)^{m-1}n^{-\lambda}$, again uniformly (as recorded in the proof of Lemma 8.5, the estimates of §§4–5 hold for any fixed weight parameter). Hence, surely,

$$\big|\,\ell(s) + \lambda\log n - (m-1)\log\log n\,\big| \;\le\; \tfrac{\beta}{2}\,\|\xi_n\|_\infty^2 + C \qquad \big(s \in [-\tfrac14,\tfrac14],\ n \ge n_0\big).$$

By convexity, for $\sigma \in [-\frac18,\frac18]$ and step $\rho = \frac18$ (so that $\sigma-\rho \in [-\frac14, 0]$),

$$\beta\,\langle t\rangle_{n,\sigma} \;=\; -\ell'(\sigma) \;\le\; \frac{\ell(\sigma-\rho)-\ell(\sigma)}{\rho} \;\le\; 8\big(\beta\,\|\xi_n\|_\infty^2 + 2C\big),$$

the $\lambda\log n - (m-1)\log\log n$ terms cancelling in the difference. Thus $\langle t\rangle_{n,\sigma} \le \mathcal{E}_n$ for all $\sigma \in [-\frac18,\frac18]$, with $C_\star := 8 + 16C/\beta$.

Step 2: proof of (i). $\big\langle e^{\delta\beta t}\big\rangle_n = Z^{(1)}(1,-\delta)/Z^{(1)}(1,0) = e^{\ell(-\delta)-\ell(0)}$ and

$$\ell(-\delta) - \ell(0) \;=\; \beta\int_{-\delta}^{0}\langle t\rangle_{n,\sigma}\,d\sigma \;\le\; \delta\beta\,\mathcal{E}_n$$

by Step 1, since $[-\delta,0] \subseteq [-\frac18,0]$.

Step 3: proof of (ii). For $t \ge 0$ and $\delta > 0$, maximizing $t \mapsto t^r e^{-\delta\beta t}$ gives $t^r \le \big(\frac{r}{e\delta\beta}\big)^r e^{\delta\beta t}$ pointwise; hence, by (i), for every $\delta \in (0,\frac18]$,

$$\big\langle t^r\big\rangle_n \;\le\; \Big(\frac{r}{e\delta\beta}\Big)^r e^{\delta\beta\,\mathcal{E}_n}.$$

If $r \le \beta\mathcal{E}_n/8$, choose $\delta\beta = r/\mathcal{E}_n \le \beta/8$: the bound is $(\mathcal{E}_n/e)^r e^r = \mathcal{E}_n^r$. Otherwise choose $\delta\beta = \beta/8$: the bound is $(8r/(e\beta))^r e^{\beta\mathcal{E}_n/8} < (8r/(e\beta))^r e^r = (8r/\beta)^r$. In either case $\langle t^r\rangle_n \le (\mathcal{E}_n + 8r/\beta)^r$. $\square$

Corollary 8.19 (Tilted moment envelopes; cf. [1, Lem. 6.5])**.** *For $x \in \mathbb{R}^N$ and $|\sigma| \le 1$ let $\langle\cdot\rangle_n^{\sigma,x}$ denote the renormalized posterior tilted by $e^{-\sigma f(x,\cdot)}$, i.e. with weight $e^{-\beta t+\beta\sqrt{t}\,\xi_n(u)-\sigma\sqrt{t/n}\,a(x,u)}$, and set*

$$\Xi_n(x) \;:=\; \frac{M(x)\,\big(1+\|\xi_n\|_\infty\big)}{\sqrt{n}}.$$

There are deterministic n_0, c_0 and, for each $r > 0$, C_r, such that surely, for all $n \ge n_0$, uniformly in $|\sigma| \le 1$ and $x \in \mathbb{R}^N$,

$$\big\langle t^{\,r}\big\rangle_n^{\sigma,x} \;\le\; C_r\,\big(1+\|\xi_n\|_\infty^2\big)^r\,\exp\Big(c_0\big(\Xi_n(x)+\Xi_n(x)^2\big)\Big).$$

Proof. Write $g := \sigma\sqrt{t/n}\,a(x,\cdot)$, so $|g| \le c\,M(x)\sqrt{t}/\sqrt{n}$ by the Cauchy-estimate envelope $|a(x,u)| \le c\,M(x)$ from the proof of Lemma 6.1. For the denominator, Jensen's inequality and Theorem 8.18(ii) with $r=1$ give

$$\big\langle e^{-g}\big\rangle_n \;\ge\; e^{-\langle g\rangle_n} \;\ge\; \exp\Big(-\frac{cM(x)}{\sqrt{n}}\,\langle t\rangle_n^{1/2}\Big) \;\ge\; \exp\big(-c'\,\Xi_n(x)\big).$$

For the numerator, by Cauchy–Schwarz and $e^{-g} \le e^{cM\sqrt{t}/\sqrt{n}}$,

$$\big\langle t^r e^{-g}\big\rangle_n \;\le\; \big\langle t^{2r}\big\rangle_n^{1/2}\,\big\langle e^{2cM\sqrt{t}/\sqrt{n}}\big\rangle_n^{1/2},$$

and by $\frac{2cM}{\sqrt{n}}\sqrt{t} \le \delta\beta t + \frac{c^2M^2}{\delta\beta n}$ together with Theorem 8.18(i), for every $\delta \in (0, \frac{1}{8}]$,

$$\big\langle e^{2cM\sqrt{t}/\sqrt{n}}\big\rangle_n \;\le\; \exp\Big(\frac{c^2M^2}{\delta\beta n} + \delta\beta\,\mathcal{E}_n\Big).$$

Choosing $\delta\beta = \min\big(cM/\sqrt{\mathcal{E}_n n},\, \beta/8\big)$ makes the exponent at most $2cM\sqrt{\mathcal{E}_n}/\sqrt{n}+8c^2M^2/(\beta n) \le c_1\big(\Xi_n(x)+\Xi_n(x)^2\big)$ (in the second regime $\beta\mathcal{E}_n/8 \le cM\sqrt{\mathcal{E}_n}/\sqrt{n}$). Combining the three displays with Theorem 8.18(ii) at $2r$ gives the claim. □

Remark 8.20. Theorem 8.18(ii) with $r=1$ recovers the sure envelope inside the proof of Lemma 8.17, and matches, at finite n, the limit-kernel envelopes of Corollary 8.8; the growth $\sim r^r$ of the constant in r is sharp, since already at $\eta=0$ one has $\langle\!\langle t^r\rangle\!\rangle_0 = \Gamma(\lambda{+}r)/(\Gamma(\lambda)\beta^r)$. No integration by parts at finite n is used: nonnegativity of t converts the r-th moment into an exponential insertion at the optimized rate $\delta \sim r/\mathcal{E}_n$, and the insertion's logarithm is a coupling-difference of free energies over the short interval $[-\delta, 0]$, controlled by the first-derivative mechanism. Corollary 8.19 is a coordinate-free counterpart of the tilted moment bounds of [1, Lem. 6.5] — weaker pointwise (the bound there is polynomial in $M(x)$, with no exponential factor) but sufficient: on $\{\Xi_n(x) \le 1\}$ the exponential factor is a constant, and the complementary events are controlled by polynomial moments of M and $\|\xi_n\|_\infty$. Lemma 8.21 below carries out the resulting third-order Taylor bookkeeping — the Lagrange remainder of $\theta \mapsto -\log\langle e^{-\theta f(x,\cdot)}\rangle_n$ is an interior-θ third cumulant, i.e. a tilted third absolute moment covered by Corollary 8.19 — and thereby removes the last citation from the derivation. One point deserves emphasis: on the training side the exceptional event $\{\Xi_n(X_i) > 1\}$ couples $M(X_i)$ to $\|\xi_n\|_\infty$, and a direct tail split fails at index $s=6$; the proof decouples them by leaving X_i out of ξ_n.

Lemma 8.21 (Cubic remainders)**.** *For $x \in \mathbb{R}^N$ let $\phi_x := \sqrt{t/n}\,a(x,\cdot)$, so that $\phi_x = f(x, g(\cdot))$ on M, and define the cubic remainder of the inner Bayes loss,*

$$R_n(x) \;:=\; -\log\big\langle e^{-\phi_x}\big\rangle_n \;-\; \langle\phi_x\rangle_n \;+\; \tfrac{1}{2}\Big(\langle\phi_x^2\rangle_n - \langle\phi_x\rangle_n^2\Big).$$

Under index $s=6$ there are a deterministic n_2 and a constant C, depending only on the standing data, such that for $n \ge n_2$,

$$n\,\mathbb{E}\,\mathbb{E}_X\big|R_n(X)\big| \;\le\; C\,n^{-1/2}, \qquad \mathbb{E}\Big[\sum_{i=1}^n\big|R_n(X_i)\big|\Big] \;\le\; C\,n^{-1/2}.$$

Proof. Step 1: Lagrange form and a sure bound. Fix x and set $h(\theta) := -\log\langle e^{-\theta\phi_x}\rangle_n$ for $\theta \in [0,1]$. Since $|\phi_x| \le c\,M(x)\sqrt{t/n} \le c\,M(x)\sqrt{\epsilon}$ is bounded, h is smooth, with $h(0) = 0$, $h'(0) = \langle\phi_x\rangle_n$, $h''(0) = -\big(\langle\phi_x^2\rangle_n - \langle\phi_x\rangle_n^2\big)$, and

$$h'''(\theta) \;=\; \Big\langle \big(\phi_x - \langle\phi_x\rangle_n^{\theta,x}\big)^3 \Big\rangle_n^{\theta,x},$$

the third central moment under the tilted posterior of Corollary 8.19 with $\sigma = \theta$. By Taylor's theorem, $R_n(x) = \frac{1}{6}\,h'''(\theta^*)$ for some $\theta^* = \theta^*(x, D_n) \in (0,1)$. From $(|\phi| + \langle|\phi|\rangle)^3 \le 4(|\phi|^3 + \langle|\phi|\rangle^3)$ and Jensen's inequality, $|h'''(\theta)| \;\le\; 8\,\langle|\phi_x|^3\rangle_n^{\theta,x} \;\le\; 8\,c^3 M(x)^3\, n^{-3/2}\,\langle t^{3/2}\rangle_n^{\theta,x}$, whence, surely,

$$n\,|R_n(x)| \;\le\; \tfrac{4}{3}\,c^3\, n^{-1/2}\, M(x)^3\, T_n(x), \qquad T_n(x) := \sup_{0\le\theta\le 1} \langle t^{3/2}\rangle_n^{\theta,x}. \tag{16}$$

Corollary 8.19 at $r = \frac{3}{2}$ (uniform in $\theta \in [0,1]$) and the deterministic bound $t \le n\epsilon$ on the inner region give the two estimates

$$T_n(x) \;\le\; C'\big(1 + \|\xi_n\|_\infty^2\big)^{3/2} \quad \text{on } \{\Xi_n(x) \le 1\},\; n \ge n_0, \qquad T_n(x) \;\le\; (n\epsilon)^{3/2} \;\text{ surely}, \tag{17}$$

with $C' = C_{3/2}\, e^{2c_0}$.

Step 2: generalization side. Split on $\{\Xi_n(X) \le 1\}$; by (16), (17), dropping the indicator on the good part, and $X \perp D_n$,

$$n\,\mathbb{E}\,\mathbb{E}_X\big|R_n(X)\big| \;\le\; C\,n^{-1/2}\,\mathbb{E}\big[(1{+}\|\xi_n\|_\infty^2)^{3/2}\big]\,\mathbb{E}_X[M^3] + C\,n\,\mathbb{E}\,\mathbb{E}_X\Big[M^3\,\mathbf{1}\big\{M\,(1{+}\|\xi_n\|_\infty) > \sqrt{n}\big\}\Big].$$

The first term is $O(n^{-1/2})$. For the second, condition on the sample and apply the Chebyshev bound $\mathbb{E}_X[M^3\,\mathbf{1}\{M > \tau\}] \le \mathbb{E}_X[M^6]/\tau^3$ at $\tau = \sqrt{n}/(1 + \|\xi_n\|_\infty)$:

$$n\,\mathbb{E}\,\mathbb{E}_X[\,\cdots] \;\le\; n\cdot n^{-3/2}\,\mathbb{E}\big[(1 + \|\xi_n\|_\infty)^3\big]\,\mathbb{E}_X[M^6] \;=\; O(n^{-1/2}).$$

Step 3: training side, good part. By (16), (17), and Cauchy–Schwarz, with $\bar{M}_n^{(3)} := \frac{1}{n}\sum_i M(X_i)^3$,

$$\mathbb{E}\Big[\sum_i |R_n(X_i)|\,\mathbf{1}\{\Xi_n(X_i) \le 1\}\Big] \;\le\; C\,n^{-1/2}\,\mathbb{E}\Big[\bar{M}_n^{(3)}\,(1{+}\|\xi_n\|_\infty^2)^{3/2}\Big] \;\le\; C\,n^{-1/2}\big(\mathbb{E}[M^6]\big)^{1/2}\big(\mathbb{E}\big[(1{+}\|\xi_n\|_\infty^2)^3\big]\big)^{1/2},$$

using $\mathbb{E}[(\bar{M}_n^{(3)})^2] \le \mathbb{E}[M^6]$ (Jensen) and $\mathbb{E}\|\xi_n\|_\infty^6 \le$ const under $s = 6$ (proof of Lemma 7.3 run at index 6): this is $O(n^{-1/2})$.

Step 4: training side, exceptional part, by decoupling. By exchangeability and the sure fallback in (17),

$$\mathbb{E}\Big[\sum_i |R_n(X_i)|\,\mathbf{1}\{\Xi_n(X_i) > 1\}\Big] \;\le\; C\,n\,\mathbb{E}\Big[M(X_1)^3\,\mathbf{1}\big\{M(X_1)(1 + \|\xi_n\|_\infty) > \sqrt{n}\big\}\Big].$$

Here $M(X_1)$ and $\|\xi_n\|_\infty$ are *not* independent, and a direct tail split fails at index $s = 6$; we decouple. Let $\xi_n^{(1)} := n^{-1/2}\sum_{i\ge 2}\big(\mathbb{E}_X[a] - a(X_i,\cdot)\big)$, so that $\xi_n = \xi_n^{(1)} + n^{-1/2}(\mathbb{E}_X[a] - a(X_1,\cdot))$, $\xi_n^{(1)}$ is independent of X_1 and equal in law to $\sqrt{(n-1)/n}\;\xi_{n-1}$, and, by $|\mathbb{E}_X[a(X,u)]| = u^k \le \sqrt{\epsilon} \le 1$ (the standing normalization $\epsilon \in (0,1)$ of Theorem 2.2) and $|a(x,u)| \le cM(x)$,

$$\|\xi_n\|_\infty \;\le\; \|\xi_n^{(1)}\|_\infty + n^{-1/2}\big(1 + c\,M(X_1)\big).$$

Consequently

$$\big\{M(X_1)(1+\|\xi_n\|_\infty) > \sqrt{n}\big\} \;\subseteq\; \Big\{M(X_1)\big(2+\|\xi_n^{(1)}\|_\infty\big) > \tfrac{\sqrt{n}}{2}\Big\} \cup \Big\{n^{-1/2}M(X_1)\big(1+c\,M(X_1)\big) > \tfrac{\sqrt{n}}{2}\Big\}.$$

For the first event, conditioning on $\xi_n^{(1)}$ and applying the Chebyshev bound in M at $\tau = \sqrt{n}/\big(2(2+\|\xi_n^{(1)}\|_\infty)\big)$,

$$n\,\mathbb{E}\big[M(X_1)^3\,\mathbf{1}\{\cdot\}\big] \;\le\; 8\,n^{-1/2}\,\mathbb{E}\big[(2+\|\xi_n^{(1)}\|_\infty)^3\big]\,\mathbb{E}[M^6] \;=\; O(n^{-1/2}),$$

with $\sup_n \mathbb{E}\|\xi_n^{(1)}\|_\infty^3 < \infty$. The second event forces $M(X_1) \ge c_1\sqrt{n}$ once $n \ge n_2$: if $M \le 1$ its left side is at most $(1+c)\,n^{-1/2} < \sqrt{n}/2$, while if $M \ge 1$ it implies $(1+c)M^2 > n/2$. Hence

$$n\,\mathbb{E}\big[M(X_1)^3\,\mathbf{1}\{M(X_1) \ge c_1\sqrt{n}\}\big] \;\le\; n\,\frac{\mathbb{E}[M^6]}{(c_1\sqrt{n})^3} \;=\; O(n^{-1/2}).$$

Summing Steps 2–4 proves the lemma. □

8.7 The Bayes pair

Write $\rho(u,v) := \mathbb{E}_X[a(X,u)a(X,v)]$, a real analytic bounded function on $M \times M$ (the $L^2(q)$-pairing of analytic families) with $\rho = \mathbb{E}[\xi\xi]$ on $\Sigma \times \Sigma$ and $\rho(u,u) = 2$ on Σ (Theorem 2.4); and recall from the proof of Lemma 6.1 the envelope bounds $|a(x,u)| + |\nabla_u a(x,u)| \le c\,M(x)$ on M, from the Cauchy estimates.

Lemma 8.22 (Second-order reduction)**.** *Under index $s = 6$,*

$$n\,B_g = \langle t\rangle_n - \frac{1}{2}\,\mathbb{E}_X\Big[\langle t\,a(X,\cdot)^2\rangle_n - \langle\sqrt{t}\,a(X,\cdot)\rangle_n^2\Big] + \varepsilon_n^g,$$
$$n\,B_t = \langle t - \sqrt{t}\,\xi_n\rangle_n - \frac{1}{2}\,\frac{1}{n}\sum_{i=1}^n\Big[\langle t\,a(X_i,\cdot)^2\rangle_n - \langle\sqrt{t}\,a(X_i,\cdot)\rangle_n^2\Big] + \varepsilon_n^t,$$

with $\mathbb{E}|\varepsilon_n^g| + \mathbb{E}|\varepsilon_n^t| \to 0$.

Proof. By Lemma 8.27 in §8.8 below (whose proof is independent of the present lemma), every posterior in Definition 8.1 may be replaced by its inner-conditioned version at the cost of errors vanishing in mean; by (3) and (5), the inner-conditioned losses are $-\log\langle e^{-\phi_x}\rangle_n$ with $\phi_x = \sqrt{t/n}\,a(x,\cdot)$, evaluated at $x = X$ resp. $x = X_i$. Lemma 8.21 expands each such loss as $\langle\phi_x\rangle_n - \frac{1}{2}(\langle\phi_x^2\rangle_n - \langle\phi_x\rangle_n^2) + R_n(x)$. The first-order terms are *exact*: $\mathbb{E}_X[a(X,u)] = \sqrt{t/n}$ on all of M (Theorem 2.3) gives $n\,\mathbb{E}_X\langle\phi_X\rangle_n = \langle t\rangle_n$, and $\frac{1}{n}\sum_i a(X_i,u) = \sqrt{t/n} - \xi_n(u)/\sqrt{n}$ (definition (4)) gives $\sum_i\langle\phi_{X_i}\rangle_n = \langle t - \sqrt{t}\,\xi_n\rangle_n$, both by linearity, with no law-of-large-numbers error. The second-order terms are as displayed, by $n\,\mathrm{Var}_n(\phi_x) = \langle t\,a(x,\cdot)^2\rangle_n - \langle\sqrt{t}\,a(x,\cdot)\rangle_n^2$. The errors ε_n^g, ε_n^t thus collect the localization terms of Lemma 8.27 and the cubic remainders $n\,\mathbb{E}_X[R_n(X)]$, resp. $\sum_i R_n(X_i)$, of Lemma 8.21; all vanish in mean under index $s = 6$, and no estimate is imported from [1] (cf. Remark 9.3). □

Lemma 8.23 (Ingredient limits)**.** *Under index $s = 4$:*

(a) $\mathbb{E}\langle t\rangle_n \to \lambda/\beta + \nu$ *and* $\mathbb{E}\langle t - \sqrt{t}\,\xi_n\rangle_n \to \lambda/\beta - \nu$*;*

(b) $\mathbb{E}\,\mathbb{E}_X\langle t\,a(X,\cdot)^2\rangle_n \to 2(\lambda/\beta+\nu)$*, and* $\mathbb{E}\big[\frac{1}{n}\sum_i\langle t\,a(X_i,\cdot)^2\rangle_n\big]$ *has the same limit;*

(c) $\mathbb{E}\,\mathbb{E}_X\big[\langle\sqrt{t}\,a(X,\cdot)\rangle_n^2\big] \to 2\lambda/\beta+2\nu-2\nu/\beta$*, and* $\mathbb{E}\big[\frac{1}{n}\sum_i\langle\sqrt{t}\,a(X_i,\cdot)\rangle_n^2\big]$ *has the same limit.*

Proof. (a) By Proposition 8.4 with $F=t$ and $F=t-\sqrt{t}\,\xi_n$ (the latter in $\mathcal{P}_1$ with P_1-norm $\le C(1+\|\xi_n\|_{C^1})$, absorbed into Θ_n), together with the continuity of $\eta\mapsto\langle\!\langle t\rangle\!\rangle_\eta$ and $\eta\mapsto\langle\!\langle t-\sqrt{t}\,\eta\rangle\!\rangle_\eta$, the two averages converge in law to $\langle\!\langle t\rangle\!\rangle_\xi$ and $\langle\!\langle t-\sqrt{t}\,\xi\rangle\!\rangle_\xi$. They are AUI: Lemma 8.17 bounds their second moments, and Theorem 7.2(b) applies. Theorem 7.2(c) and Proposition 8.10 conclude.

(b) By Fubini, $\mathbb{E}_X\langle t\,a(X,\cdot)^2\rangle_n=\langle t\,\rho(u,u)\rangle_n$ with $\rho(u,u)$ real analytic, bounded, equal to 2 on Σ; by Proposition 8.4 this converges in law to $\langle\!\langle t\,\rho(u,u)\rangle\!\rangle_\xi=2\langle\!\langle t\rangle\!\rangle_\xi$ (the limit kernel is supported on Σ), is dominated by $\|\rho\|_\infty\langle t\rangle_n$ hence AUI, and its mean converges to $2\mathbb{E}\langle\!\langle t\rangle\!\rangle=2(\lambda/\beta+\nu)$. For the empirical version, $\frac{1}{n}\sum_i a(X_i,u)^2=\rho(u,u)-n^{-1/2}\chi_n(u)$ where χ_n is the centred empirical process of the family $a(\cdot,u)^2$; this family is $L^2(q)$-valued analytic because a is $L^4(q)$-valued analytic (Cauchy products, $\|a_\alpha a_{\alpha'}\|_2\le\|a_\alpha\|_4\|a_{\alpha'}\|_4$), so $\sup_n\mathbb{E}\|\chi_n\|_\infty^2<\infty$ by [1, Thm. 5.8] with $s=2$. Then $\big|\frac{1}{n}\sum_i\langle ta(X_i)^2\rangle_n-\langle t\rho(u,u)\rangle_n\big|=n^{-1/2}|\langle t\,\chi_n\rangle_n|\le n^{-1/2}\|\chi_n\|_\infty\langle t\rangle_n$, whose mean is $O(n^{-1/2})$ by Cauchy–Schwarz and Lemma 8.17.

(c) Fix x. The insertion $F_x:=\sqrt{t}\,a(x,\cdot)$ lies in $\mathcal{P}_{1/2}$ with $P_{1/2}(F_x)\le c\,M(x)$, so Proposition 8.4(i) gives, uniformly in x on Ω_n, $|\langle\sqrt{t}a(x)\rangle_n-\langle\!\langle\sqrt{t}a(x,\cdot)\rangle\!\rangle_{\xi_n}|\le CM(x)\Theta_n r_n'$; moreover $\langle\sqrt{t}a(x)\rangle_n^2\le\langle ta(x)^2\rangle_n\le c^2M(x)^2\langle t\rangle_n$ by Jensen. Since $\mathbb{E}_XM^2<\infty$, integrating in x shows $\mathbb{E}_X[\langle\sqrt{t}a(X)\rangle_n^2]-G(\xi_n)\to 0$ in probability, where $G(\eta):=\mathbb{E}_X\big[\langle\!\langle\sqrt{t}\,a(X,\cdot)\rangle\!\rangle_\eta^2\big]$ is continuous on $C(M)$ (dominated convergence with the same majorant). Hence $\mathbb{E}_X[\langle\sqrt{t}a(X)\rangle_n^2]\to G(\xi)$ in law; it is dominated by $c^2\mathbb{E}_X[M^2]\,\langle t\rangle_n$, hence AUI by Lemma 8.17, so its mean converges to $\mathbb{E}[G(\xi)]=\mathbb{E}\,\mathbb{E}_X\big[\langle\!\langle\sqrt{t}\,a(X,u)\rangle\!\rangle_\xi^2\big]$. This last expectation is evaluated by the Gaussian integration by parts of Theorem 8.13: $\mathbb{E}\,\mathbb{E}_X[\langle\!\langle\sqrt{t}a\rangle\!\rangle^2]=\frac{2}{\beta^2}\big(\lambda\beta+(\beta^2-\beta)\nu\big)=2\lambda/\beta+2\nu-2\nu/\beta$. For the empirical version, use the two-replica identity

$$\frac{1}{n}\sum_i\langle\sqrt{t}\,a(X_i,\cdot)\rangle_n^2=\Big\langle\sqrt{t}\,\sqrt{t'}\,\frac{1}{n}\sum_i a(X_i,u)\,a(X_i,v)\Big\rangle_n^{(2)}=\big\langle\sqrt{t}\sqrt{t'}\,\rho(u,v)\big\rangle_n^{(2)}-n^{-1/2}\big\langle\sqrt{t}\sqrt{t'}\,\tilde\chi_n(u,v)\big\rangle_n^{(2)},$$

where $\tilde\chi_n$ is the centred empirical process on $M\times M$ of the $L^2(q)$-valued analytic family $a(\cdot,u)a(\cdot,v)$, with $\sup_n\mathbb{E}\|\tilde\chi_n\|_\infty^2<\infty$ as in (b). The correction term has mean $O(n^{-1/2})$ since $|\langle\sqrt{t}\sqrt{t'}\,\tilde\chi_n\rangle^{(2)}|\le\|\tilde\chi_n\|_\infty\langle\sqrt{t}\rangle_n^2\le\|\tilde\chi_n\|_\infty\langle t\rangle_n$. The main term converges in law to $\langle\!\langle\sqrt{t}\sqrt{t'}\rho(u,v)\rangle\!\rangle_\xi^{(2)}$ by Proposition 8.4(iii), is dominated by $\|\rho\|_\infty\langle t\rangle_n$ hence AUI, and its mean converges to $\mathbb{E}\langle\!\langle\sqrt{t}\sqrt{t'}\rho(u,v)\rangle\!\rangle_\xi^{(2)}=\mathbb{E}\,\mathbb{E}_X\big[\langle\!\langle\sqrt{t}a(X,u)\rangle\!\rangle_\xi^2\big]$ by Fubini, the same limit as before. $\square$

Theorem 8.24 (Bayes errors)**.** *Under index $s=6$,*

$$n\,\mathbb{E}[B_g]\;\longrightarrow\;\frac{\lambda}{\beta}+\nu-\frac{\nu}{\beta},\qquad n\,\mathbb{E}[B_t]\;\longrightarrow\;\frac{\lambda}{\beta}-\nu-\frac{\nu}{\beta}.$$

Proof. Combine Lemmas 8.22 and 8.23:

$$n\mathbb{E}[B_g]\to\Big(\frac{\lambda}{\beta}+\nu\Big)-\frac{1}{2}\Big[2\Big(\frac{\lambda}{\beta}+\nu\Big)-\Big(\frac{2\lambda}{\beta}+2\nu-\frac{2\nu}{\beta}\Big)\Big]=\frac{\lambda}{\beta}+\nu-\frac{\nu}{\beta},$$

and likewise $n\mathbb{E}[B_t]\to(\lambda/\beta-\nu)-\nu/\beta$. $\square$

Remark 8.25. Subtracting the limits in Lemma 8.23(b) and (c) gives

$$\nu(\beta) \;=\; \frac{\beta}{2}\,\mathbb{E}\,\mathbb{E}_X\Big[\langle\!\langle t\,a(X,u)^2\rangle\!\rangle_\xi - \langle\!\langle \sqrt{t}\,a(X,u)\rangle\!\rangle_\xi^2\Big] \;\ge\; 0,$$

the mean posterior variance form of the singular fluctuation, in agreement with [1, Rem. 5.9(2)]; in particular $\nu \ge 0$. Consequently the leading coefficients in Theorem 8.2 obey, for every $\beta > 0$,

$$\mathbb{E}[B_t] \;\le\; \mathbb{E}[G_t] \;\le\; \mathbb{E}[G_g] \qquad \text{and} \qquad \mathbb{E}[B_t] \;\le\; \mathbb{E}[B_g] \;\le\; \mathbb{E}[G_g]$$

up to $o(1/n)$, the successive gaps being ν/β, 2ν and 2ν, ν/β (all times $1/n$). The remaining comparison changes sign at $\beta = \frac{1}{2}$: since $\mathbb{E}[B_g] - \mathbb{E}[G_t] = \nu(2 - 1/\beta)/n + o(1/n)$, one has $\mathbb{E}[G_t] \le \mathbb{E}[B_g]$ for $\beta \ge \frac{1}{2}$ but the reverse inequality for $\beta < \frac{1}{2}$ whenever $\nu > 0$. We also note that two of these orderings need no asymptotics at all: $B_t \le G_t$ and $0 \le B_g \le G_g$ hold surely at every n, by Jensen's inequality applied to $-\log\langle e^{-f}\rangle$ ([1, Lem. 6.1]).

8.8 Random-variable versions, the empirical variance, and WAIC

The results so far are statements about expectations. In [1, §6.3] the central statements are instead at the level of random variables: the quartet converges jointly in law, and certain combinations of its entries degenerate in probability — and it is the latter, being observable, that make the theory usable in practice ([1, Rem. 6.8(2)]). This subsection derives those statements; every estimate needed has already been established above, so the proofs are assemblies.

For $\eta \in C(M)$ define the limit functionals (cf. [1, Def. 6.9])

$$G_g^*(\eta) := \langle\!\langle t\rangle\!\rangle_\eta, \qquad G_t^*(\eta) := \langle\!\langle t - \sqrt{t}\,\eta(u)\rangle\!\rangle_\eta, \qquad B_g^*(\eta) := \tfrac{1}{2}\,\mathbb{E}_X\big[\langle\!\langle \sqrt{t}\,a(X,u)\rangle\!\rangle_\eta^2\big],$$

and $B_t^*(\eta) := B_g^*(\eta) + G_t^*(\eta) - G_g^*(\eta)$. In [1, Def. 6.9] this last identity is the *definition* of the Bayes training limit; here it will be derived. Define further, following [1, Def. 6.6], the *empirical variance* and its renormalized inner version

$$V \;:=\; \sum_{i=1}^n\Big(\big\langle f(X_i,\cdot)^2\big\rangle - \big\langle f(X_i,\cdot)\big\rangle^2\Big), \qquad W_n \;:=\; \frac{1}{n}\sum_{i=1}^n\Big(\big\langle t\,a(X_i,\cdot)^2\big\rangle_n - \big\langle \sqrt{t}\,a(X_i,\cdot)\big\rangle_n^2\Big);$$

by $f(x,g(u)) = a(x,u)\sqrt{t/n}$, the inner-restricted posterior variances of the $f(X_i,\cdot)$ sum to exactly W_n. Note that V, like B_t, G_t, is observable.

Lemma 8.26 (Localization)**.** *Let $\Pi_n := \langle \mathbf{1}\{K > \epsilon\}\rangle$ be the posterior mass of the outer region and $\bar{M}_n^{(p)} := \frac{1}{n}\sum_i M(X_i)^p$.*

- *(i) Surely for $n \ge n_0$, $\Pi_n \le C\,n^\lambda e^{-n\beta\epsilon/2}\,e^{\beta V_n}$, with V_n as in Section 7; in particular $n^p\,\Pi_n \to 0$ in probability for every fixed p.*
- *(ii) $nG_g - \langle t\rangle_n \to 0$, $nG_t - \langle t - \sqrt{t}\,\xi_n\rangle_n \to 0$, and $V - W_n \to 0$, in probability.*
- *(iii) Under index $s = 6$, moreover $\mathbb{E}\,|V - W_n| \to 0$.*

Proof. (i) By the outer-region display in the proof of Theorem 6.4, $Z_n^{(2)} \le e^{-n\beta\epsilon/2}\exp\big(\frac{\beta}{2}\sup_{K>\epsilon}\psi_n^2\big)$, while the proof of Lemma 7.4(i) gives $Z_n^0 \ge c_1\,n^{-\lambda}(\log n)^{m-1}e^{-\beta\|\xi_n\|_\infty^2/2}$ for $n \ge n_0$; divide, and absorb both exponents into $e^{\beta V_n}$. Since V_n is bounded in probability (Theorem 2.4), $n^p \Pi_n \to 0$ in probability.

(ii) Splitting the posterior at $\{K \le \epsilon\}$, $nG_g = (1-\Pi_n)\langle t\rangle_n + n\big\langle K\,\mathbf{1}\{K > \epsilon\}\big\rangle$, hence $|nG_g - \langle t\rangle_n| \le \Pi_n|\langle t\rangle_n| + nK_{\max}\Pi_n$; the first term vanishes in probability because $\langle t\rangle_n$ is bounded in probability (Lemma 8.17), the second by (i). For nG_t, replace $nK_{\max}$ by $n\sup_W |K_n| \le n\bar{M}_n^{(1)}$ and use $\bar{M}_n^{(1)} = O_P(1)$. For the third statement: for h with $\|h\|_\infty \le M(X_i)^p$ one has $|\langle h\rangle - \langle h\rangle_n| \le \Pi_n|\langle h\rangle_n| + |\langle h\,\mathbf{1}\{K > \epsilon\}\rangle| \le 2\,\Pi_n M(X_i)^p$; applying this to $f(X_i,\cdot)^2$ and to $f(X_i,\cdot)$ (the latter multiplied by $|\langle f\rangle + \langle f\rangle_n| \le 2M(X_i)$) bounds the i-th variance discrepancy by $6\,\Pi_n M(X_i)^2$, so $|V - W_n| \le 6\,n\,\Pi_n\,\bar{M}_n^{(2)} \to 0$ in probability, by (i) and the law of large numbers for $\bar{M}_n^{(2)}$.

(iii) Split on $E_n := \{V_n \le n\epsilon/4\}$. On E_n, (i) gives $\Pi_n \le Cn^\lambda e^{-n\beta\epsilon/4}$, so $\mathbb{E}\big[|V - W_n|\,\mathbf{1}_{E_n}\big] \le 6C\,n^{\lambda+1}e^{-n\beta\epsilon/4}\,\mathbb{E}[M^2] \to 0$. Off E_n, use the sure bound $|V - W_n| \le V + W_n \le 2n\bar{M}_n^{(2)}$ (each posterior variance lies in $[0, M(X_i)^2]$) and Cauchy–Schwarz:

$$\mathbb{E}\big[2n\bar{M}_n^{(2)}\mathbf{1}_{E_n^c}\big] \;\le\; 2n\left(\mathbb{E}[M^4]\right)^{1/2}\,\Pr(E_n^c)^{1/2} \;=\; O(n^{-1/2}),$$

since $\mathbb{E}[(\bar{M}_n^{(2)})^2] \le \mathbb{E}[M^4] < \infty$ (Jensen; $M \in L^6$) and $\Pr(E_n^c) \le \mathbb{E}[V_n^3]\,(4/(n\epsilon))^3 = O(n^{-3})$ by Chebyshev, the sixth moments $\mathbb{E}\|\xi_n\|_\infty^6$, $\mathbb{E}\sup_{K\ge\epsilon}\psi_n^6 \le$ const being available under $s = 6$ by the proof of Lemma 7.3 run at index 6. $\square$

Lemma 8.27 (Localization in mean for the Bayes losses)**.** *For $x \in \mathbb{R}^N$ let*

$$\mathcal{L}_n(x) := -\log\big\langle e^{-f(x,\cdot)}\big\rangle, \qquad \mathcal{L}_n^{(1)}(x) := -\log\big\langle e^{-f(x,\cdot)}\,\big|\,K \le \epsilon\big\rangle,$$

the Bayes losses of the full and of the inner-conditioned posterior; by (3) *and* (5), $\mathcal{L}_n^{(1)}(x) = -\log\langle e^{-\sqrt{t/n}\,a(x,\cdot)}\rangle_n$, *and the conditioning is well defined since the inner region carries positive posterior mass surely. Under index $s = 6$,*

$$n\,\mathbb{E}\,\mathbb{E}_X\big|\mathcal{L}_n(X) - \mathcal{L}_n^{(1)}(X)\big| \;\longrightarrow\; 0, \qquad \mathbb{E}\sum_{i=1}^n\big|\mathcal{L}_n(X_i) - \mathcal{L}_n^{(1)}(X_i)\big| \;\longrightarrow\; 0$$

(the first limit needs only $s = 4$). In particular nB_g and nB_t may be computed, up to errors vanishing in mean, with every posterior average in Definition 8.1 replaced by its inner-conditioned version.

Proof. Write $\Delta(x) := \mathcal{L}_n(x) - \mathcal{L}_n^{(1)}(x)$, $D(x) := \langle e^{-f(x,\cdot)} \mid K \le \epsilon\rangle$ and $R(x) := \langle e^{-f(x,\cdot)}\,\mathbf{1}\{K > \epsilon\}\rangle \ge 0$. The mixture decomposition of the posterior at $\{K \le \epsilon\}$ gives

$$\big\langle e^{-f(x,\cdot)}\big\rangle = (1 - \Pi_n)\,D(x) + R(x), \qquad \Delta(x) = -\log\Big[(1 - \Pi_n) + \frac{R(x)}{D(x)}\Big].$$

Three sure bounds. First, $|f(x,w)| \le M(x)$ for all $w \in W$, so every posterior average of $e^{-f(x,\cdot)}$ lies in $[e^{-M(x)}, e^{M(x)}]$ and both losses lie in $[-M(x), M(x)]$:

$$|\Delta(x)| \;\le\; 2\,M(x).$$

Second, dropping $R \ge 0$ gives $\Delta(x) \le -\log(1 - \Pi_n)$, while $\log y \le y - 1$ gives $\Delta(x) \ge \Pi_n - R(x)/D(x) \ge -R(x)/D(x)$; hence, whenever $\Pi_n \le \frac{1}{2}$,

$$|\Delta(x)| \;\le\; 2\,\Pi_n + \frac{R(x)}{D(x)}.$$

Third, $R(x) \le e^{M(x)}\Pi_n$ and $D(x) \ge e^{-M(x)}$, so the second bound yields

$$|\Delta(x)| \;\le\; 3\,e^{2M(x)}\,\Pi_n \qquad \text{whenever } \Pi_n \le \tfrac{1}{2}.$$

Let $E_n := \{V_n \le n\epsilon/4\}$ as in Lemma 8.26(iii), so that on E_n, for $n \ge n_0$, Lemma 8.26(i) gives $\Pi_n \le p_n := C\,n^{\lambda}e^{-n\beta\epsilon/4}$, with $p_n \le \frac{1}{2}$ for $n \ge n_1$; and $\Pr(E_n^c) \le \mathbb{E}[V_n^3]\,\big(4/(n\epsilon)\big)^3 = O(n^{-3})$ by the sixth moments available at $s = 6$ (proof of Lemma 8.26(iii)). Set $\tau_n := n\beta\epsilon/16$. On $E_n \cap \{M(x) \le \tau_n\}$ the third bound gives, for $n \ge n_1$,

$$|\Delta(x)| \;\le\; 3\,e^{2\tau_n}p_n \;=\; 3C\,n^{\lambda}\,e^{-n\beta\epsilon/8} \;=:\; q_n,$$

a deterministic, superpolynomially small quantity: the outer posterior mass on E_n is small enough to beat the uncontrolled factor $e^{2M(x)}$ for every $M(x)$ of order n. Elsewhere use the sure envelope. Thus, surely, for $n \ge n_1$,

$$|\Delta(x)| \;\le\; q_n \;+\; 2M(x)\,\mathbf{1}\{M(x) > \tau_n\} \;+\; 2M(x)\,\mathbf{1}_{E_n^c}\,.$$

For the generalization loss, X is independent of the sample, so

$$n\,\mathbb{E}\,\mathbb{E}_X\big|\Delta(X)\big| \;\le\; n\,q_n + 2n\,\mathbb{E}_X\big[M\,\mathbf{1}\{M > \tau_n\}\big] + 2n\,\mathbb{E}_X[M]\,\Pr(E_n^c) \;=\; o(1)+O(n^{-4})+O(n^{-2}),$$

using $\mathbb{E}_X[M\,\mathbf{1}\{M > \tau_n\}] \le \mathbb{E}[M^6]/\tau_n^5$. For the training loss,

$$\mathbb{E}\sum_{i=1}^{n}\big|\Delta(X_i)\big| \;\le\; n\,q_n \;+\; 2n\,\mathbb{E}\big[M\,\mathbf{1}\{M > \tau_n\}\big] \;+\; 2\,\mathbb{E}\Big[\sum_i M(X_i)\,\mathbf{1}_{E_n^c}\Big],$$

and the last term is at most $2\big(\mathbb{E}[(\sum_i M(X_i))^2]\big)^{1/2}\,\Pr(E_n^c)^{1/2} = O(n)\cdot O(n^{-3/2}) = O(n^{-1/2})$ by Cauchy–Schwarz; here the sixth moments are genuinely used, fourth moments giving only $O(1)$. (For the generalization loss, fourth moments and $M \in L^4$ suffice in every term, whence the parenthetical in the statement.) $\square$

Theorem 8.28 (The quartet in law; cf. [1, Thm. 6.8]). *(1) Under index $s = 4$, $(nG_g,\, nG_t) \to (G_g^*(\xi),\, G_t^*(\xi))$ jointly in law; under index $s = 6$ the full quartet converges jointly,*

$$\big(nB_g,\; nB_t,\; nG_g,\; nG_t\big) \;\longrightarrow\; \big(B_g^*(\xi),\; B_t^*(\xi),\; G_g^*(\xi),\; G_t^*(\xi)\big) \quad \textit{in law.}$$

(2) Under index $s = 6$, $n\,(B_g - B_t - G_g + G_t) \to 0$ in probability.

Proof. By Lemma 8.26(ii) and Proposition 8.4(i) (with $F = t$ and $F = t - \sqrt{t}\,\xi_n$, as in Lemma 8.23(a)), $nG_g = G_g^*(\xi_n) + o_P(1)$ and $nG_t = G_t^*(\xi_n) + o_P(1)$; continuity of $\eta \mapsto (G_g^*, G_t^*)(\eta)$ (Proposition 8.4(ii)), the continuous mapping theorem and Slutsky give the Gibbs part. For the Bayes part, Lemma 8.22 and the exact Fubini identities from the proof of Lemma 8.23 give

$$nB_g \;=\; \langle t\rangle_n - \tfrac{1}{2}\big\langle t\,\rho(u,u)\big\rangle_n + \tfrac{1}{2}\big\langle \sqrt{t}\sqrt{t'}\,\rho(u,v)\big\rangle_n^{(2)} + \varepsilon_n^g,$$

while nB_t has the same representation with first term $\langle t - \sqrt{t}\,\xi_n\rangle_n$, second and third terms differing by the $O_P(n^{-1/2})$ empirical-process corrections computed in the proofs of Lemma 8.23(b),(c), and error ε_n^t. Since $\mathbb{E}|\varepsilon_n^{g,t}| \to 0$ (Lemma 8.22), both errors vanish in probability. By Proposition 8.4(i),(iii) the averages converge, jointly with the Gibbs pair, to $\langle\!\langle t\rangle\!\rangle_\xi$,

$\langle\!\langle t\,\rho(u,u)\rangle\!\rangle_\xi = 2G_g^*(\xi)$ and $\langle\!\langle \sqrt{t}\sqrt{t'}\rho(u,v)\rangle\!\rangle_\xi^{(2)} = \mathbb{E}_X\big[\langle\!\langle \sqrt{t}\,a(X,u)\rangle\!\rangle_\xi^2\big] = 2B_g^*(\xi)$ (using $\rho=2$ on Σ, and the continuity in η of the limit functionals established in the proofs of Proposition 8.4(ii) and Lemma 8.23(c)). Assembling, $nB_g \to B_g^*(\xi)$ and $nB_t \to G_t^* - G_g^* + B_g^* = B_t^*$ at ξ, jointly; the relation defining B_t^* is thus derived, not imposed. (2) In $n(B_g - B_t) - n(G_g - G_t)$ the first-order terms cancel against the Gibbs representations exactly, leaving only the $O_P(n^{-1/2})$ corrections, the localization errors of Lemma 8.26(ii), and $\varepsilon_n^g - \varepsilon_n^t$. □

Theorem 8.29 (Equations of states in probability; cf. [1, Thm. 6.9, Cor. 6.3]). *(1) Surely, for every $\eta \in C(M)$: $G_g^*(\eta) + G_t^*(\eta) = 2\lambda/\beta$.*

(2) Under index $s=4$, $nG_g + nG_t - 2\lambda/\beta \to 0$ in probability.

(3) Under index $s=6$, $nB_g - nB_t + 2nG_t - 2\lambda/\beta \to 0$ in probability.

Proof. (1) is Lemma 8.7: $G_g^* + G_t^* = 2\langle\!\langle t\rangle\!\rangle_\eta - \langle\!\langle \sqrt{t}\,\eta\rangle\!\rangle_\eta = 2\lambda/\beta$. (2) By the Gibbs representations in the proof of Theorem 8.28, $nG_g + nG_t = G_g^*(\xi_n) + G_t^*(\xi_n) + o_P(1)$, and the bracket equals $2\lambda/\beta$ surely by (1). (3) Add Theorem 8.28(2) to (2). □

Theorem 8.30 (The empirical variance; cf. [1, Def. 6.6, Thm. 6.10]). *Under index $s=6$:*

(1) $V - 2\,(nG_t - nB_t) \to 0$ and $V - 2\,(nG_g - nB_g) \to 0$ in probability;

(2) $V \to 2\,(G_t^* - B_t^*)(\xi) = 2\,(G_g^* - B_g^*)(\xi)$ in law, and $nB_g + nB_t + V - 2\lambda/\beta \to 0$ in probability;

(3) $\mathbb{E}[V] \to 2\nu(\beta)/\beta$; under index $s=4$ already $\mathbb{E}[W_n] \to 2\nu(\beta)/\beta$.

Proof. (1) By Lemma 8.22 and Lemma 8.26(ii), $2(nG_t - nB_t) = W_n - 2\varepsilon_n^t + o_P(1)$, so the first relation reduces to $V - W_n \to 0$ (Lemma 8.26(ii)) and $\varepsilon_n^t \to 0$. Likewise $2(nG_g - nB_g) = \big[\langle t\rho(u,u)\rangle_n - \langle \sqrt{t}\sqrt{t'}\rho(u,v)\rangle_n^{(2)}\big] - 2\varepsilon_n^g + o_P(1)$, and the bracket differs from W_n by the $O_P(n^{-1/2})$ corrections of Lemma 8.23(b),(c). (2) Combine (1) with Theorem 8.28(1); the identity $G_t^* - B_t^* = G_g^* - B_g^*$ is immediate from the definition of B_t^*. For the second statement, sum the two relations in (1) and insert Theorem 8.29(2). (3) By Lemma 8.23(b),(c) (empirical versions), $\mathbb{E}[W_n] \to 2(\lambda/\beta + \nu) - (2\lambda/\beta + 2\nu - 2\nu/\beta) = 2\nu/\beta$, under $s=4$; then $\mathbb{E}[V] \to 2\nu/\beta$ by Lemma 8.26(iii). □

Corollary 8.31 (Estimation identities; cf. [1, Cor. 6.2, Cor. 6.3, Rem. 6.8(2), eqs. (6.31)–(6.32)]). *Under index $s=6$,*

$$\mathbb{E}[B_g] \;=\; \mathbb{E}[B_t] + \frac{\beta}{n}\,\mathbb{E}[V] + o\Big(\frac{1}{n}\Big), \qquad \mathbb{E}[G_g] \;=\; \mathbb{E}[G_t] + \frac{\beta}{n}\,\mathbb{E}[V] + o\Big(\frac{1}{n}\Big),$$

$$\mathbb{E}[B_g] - \mathbb{E}[B_t] + 2\,\mathbb{E}[G_t] \;=\; \frac{2\lambda}{\beta n} + o\Big(\frac{1}{n}\Big),$$

and at $\beta = 1$, $\mathbb{E}[B_g] = \lambda/n + o(1/n)$, whatever the value of ν. Since B_t, G_t and V are observable, the first display estimates both generalization errors from training quantities alone; equivalently, the matrix identity of [1, Cor. 6.2] holds for the four limits of Theorem 8.2.

Proof. Immediate from Theorem 8.2 and $\mathbb{E}[V] \to 2\nu/\beta$ (Theorem 8.30), by $2\nu = \beta\cdot(2\nu/\beta)$; the third display and the $\beta=1$ case follow by substituting the four coefficients. □

The subsection's title is earned by the following application ([1, §8.3]), which we now make explicit. Assume the entropy $S = -\int q \log q \, dx$ is finite, and recall $S_n = -\frac{1}{n}\sum_i \log q(X_i)$ from §2. In practice one works with *losses* rather than errors:

$$BL_g := -\mathbb{E}_X \log\bigl\langle p(X \mid w)\bigr\rangle, \qquad BL_t := -\frac{1}{n}\sum_{i=1}^{n} \log\bigl\langle p(X_i \mid w)\bigr\rangle,$$

$$GL_g := -\bigl\langle \mathbb{E}_X \log p(X \mid w)\bigr\rangle, \qquad GL_t := -\Bigl\langle \tfrac{1}{n}\sum_{i=1}^{n} \log p(X_i \mid w)\Bigr\rangle,$$

so that $BL_g = B_g + S$, $BL_t = B_t + S_n$, $GL_g = G_g + S$, $GL_t = G_t + S_n$. The generalization losses are the quantities one wants; the training losses are computable from posterior samples with no knowledge of q, and $GL_t - BL_t = G_t - B_t$ is observable. Following [1, §8.3], the *widely applicable information criteria* are the observable estimators

$$\mathrm{WAIC}_1 \;:=\; BL_t + 2\beta\,\bigl(GL_t - BL_t\bigr), \qquad \mathrm{WAIC}_2 \;:=\; GL_t + 2\beta\,\bigl(GL_t - BL_t\bigr)$$

of the Bayes and Gibbs generalization losses respectively.

Corollary 8.32 (WAIC; cf. [1, §8.3, eqs. (8.4)–(8.8)])**.** *Under index $s = 6$ and finite entropy,*

$$\mathbb{E}\bigl[BL_g\bigr] = \mathbb{E}\bigl[\mathrm{WAIC}_1\bigr] + o\Bigl(\frac{1}{n}\Bigr), \qquad \mathbb{E}\bigl[GL_g\bigr] = \mathbb{E}\bigl[\mathrm{WAIC}_2\bigr] + o\Bigl(\frac{1}{n}\Bigr),$$

and, pathwise,

$$\mathrm{WAIC}_1 = BL_t + \frac{\beta}{n}\,V + o_P\Bigl(\frac{1}{n}\Bigr), \qquad \mathrm{WAIC}_2 = GL_t + \frac{\beta}{n}\,V + o_P\Bigl(\frac{1}{n}\Bigr).$$

Proof. Since $\mathbb{E}[S_n] = S$, subtracting entropies converts the first display into $\mathbb{E}[B_g] - \mathbb{E}[B_t] = 2\beta(\mathbb{E}[G_t] - \mathbb{E}[B_t]) + o(1/n)$ and $\mathbb{E}[G_g] - \mathbb{E}[G_t] = 2\beta(\mathbb{E}[G_t] - \mathbb{E}[B_t]) + o(1/n)$, which is Corollary 8.3. For the second display, $2\beta(GL_t - BL_t) = \frac{\beta}{n}\cdot 2\,(nG_t - nB_t) = \frac{\beta}{n}\bigl(V + o_P(1)\bigr)$ by Theorem 8.30(1). $\square$

Corollary 8.32 is what makes the theory of this section usable: it estimates a generalization loss from training quantities alone, by a formula containing no model-specific constant — neither (λ, m) nor ν appears — valid for arbitrary (q, p, φ) satisfying the fundamental conditions, regular or singular. Within this note its logical support is exactly the chain assembled above: the Gibbs pair by tilted free energies (Theorem 8.15), the Bayes pair by the second-order reduction with the internal envelopes (Theorems 8.24, 8.18, Lemma 8.21) and the Gaussian integration by parts (Theorem 8.13), the equations of states in mean (Corollary 8.3) for unbiasedness, and the localization and empirical-variance relations (Lemmas 8.26, 8.27, Theorem 8.30) for the observable V-form. In a regular model $\nu = d/2$ ([1, Rem. 6.11]), so the correction $\frac{\beta}{n}\mathbb{E}[V] \to 2\nu/n = d/n$ recovers Akaike's penalty; [1, Rem. 8.3(4), eq. (8.9)] relates the $\beta \to \infty$ limit to AIC proper, and [1, Rem. 8.3(1)] discusses the unrealizable case, which lies outside our standing assumption $W_0 \neq \emptyset$.

The precise sense in which the criteria estimate, and the precise sense in which they fluctuate, are both governed by a degeneracy of the limit quartet which the results above make visible.

Proposition 8.33 (Two degrees of freedom). *Let $W := \langle\!\langle \sqrt{t}\,\xi(u)\rangle\!\rangle_\xi$. Almost surely,*

$$G_g^* \;=\; \frac{\lambda}{\beta} + \frac{W}{2}, \qquad G_t^* \;=\; \frac{\lambda}{\beta} - \frac{W}{2}, \qquad B_t^* \;=\; B_g^* - W,$$

so the limit quartet $(B_g^*, B_t^*, G_g^*, G_t^*)$, and the limit $2(G_g^* - B_g^*)$ of V, are affine functions of the pair (W, B_g^*). Moreover $|W|$ and B_g^* are bounded surely by $C(1 + \|\xi\|_\infty^2)$, so all their moments are finite already under index $s = 2$, with $\mathbb{E}[W] = 2\nu$ and $\mathbb{E}[B_g^*] = \lambda/\beta + \nu - \nu/\beta$. In particular

$$\mathrm{Var}(G_g^*) \;=\; \mathrm{Var}(G_t^*) \;=\; -\,\mathrm{Cov}(G_g^*, G_t^*) \;=\; \tfrac{1}{4}\,\mathrm{Var}(W)$$

— the Gibbs limits are perfectly anticorrelated — and every second moment of the five limits is determined by the three numbers $\mathrm{Var}(W)$, $\mathrm{Var}(B_g^*)$, $\mathrm{Cov}(W, B_g^*)$.

Proof. The first two identities are Lemma 8.7 at $\eta = \xi$, which holds surely; the third is the relation derived in Theorem 8.28. The sure bounds are Corollary 8.8 together with $B_g^* \leq \frac{1}{2}\,\mathbb{E}_X\big[\langle\!\langle t\, a(X,u)^2\rangle\!\rangle_\xi\big] \leq \frac{c^2}{2}\,\mathbb{E}_X[M^2]\,\langle\!\langle t\rangle\!\rangle_\xi$ (Jensen and the envelope $|a| \leq cM$), and the moments are finite by Fernique's theorem [7]. The means are Definition 8.9 and Theorem 8.2. □

Remark 8.34 (Unbiased, not pathwise exact). The relations of Corollaries 8.31 and 8.32 are statements in mean, and the mean is the right level, for two distinct reasons. First, as an estimator of the loss BL_g, the criterion carries the empirical-entropy fluctuation: $\mathrm{WAIC}_1 - BL_g = (S_n - S) + O_P(1/n)$, and $S_n - S = O_P(n^{-1/2})$ dominates every $1/n$ effect; but this term does not involve (p, φ), is common to all models evaluated on the same data, and cancels from comparisons, so it is harmless for selection. Second, at the error level, where the entropies cancel exactly, the criterion is unbiased to $o(1/n)$ but not pathwise exact: by Theorem 8.28 and Proposition 8.33,

$$n\,\big(B_g - B_t - 2\beta\,(G_t - B_t)\big) \;\longrightarrow\; (1-\beta)\,W + 2\beta\,B_g^* - 2\lambda \qquad \text{in law},$$

a limit with mean zero but nondegenerate in general; the same limit governs $n(B_g - B_t) - \beta V$, by Theorem 8.30(1). At $\beta = 1$ the limit is $2\,(B_g^* - \lambda)$: twice the fluctuation of the Bayes limit about its mean, with variance $4\,\mathrm{Var}(B_g^*)$. Thus WAIC-differences between models, multiplied by n, converge to nondegenerate random variables rather than constants, and WAIC-based selection is not consistent — exactly as for AIC in regular models; see [1, Rem. 8.3(5)], whose statement is the mean-level shadow of this display, as [1, Rem. 6.9] is of Proposition 8.33. The three second moments of Proposition 8.33 are, after $\nu(\beta)$, the natural next invariants of the triple (Σ, μ, ρ); computing or bounding them — by the calculus of Remark 8.14 — is the beginning of a second-order theory that we do not pursue here. The combinations that do degenerate pathwise are exactly those of Theorems 8.28(2), 8.29(2),(3) and 8.30(1),(2).

9 Remarks

Remark 9.1 (Invariance, and the real log canonical threshold). The pair (λ, m) is intrinsic to $(K, \varphi\, dw)$ on W: it is read off the zeta function downstairs (§2), before any resolution is chosen. When $\varphi > 0$ on a neighborhood of W_0, λ is the *real log canonical threshold* of K (along W_0) and m its multiplicity; in general (λ, m) is the real log canonical threshold data of the pair $(K, \varphi\, dw)$, the zeros of φ_2 entering through the multi-indices h.

By contrast, μ, Σ, and the process ξ live on M and depend on the chosen resolution. The pushforward $g_*\mu$ is a canonical positive measure on W_0: by (3), for $\Psi \in C(W_\epsilon)$,

$$(g_*\mu)[\Psi] \;=\; \mu[\Psi \circ g] \;=\; \lim_{\sigma \to -\lambda^+} (\sigma + \lambda)^m \int_{W_\epsilon} K(w)^\sigma \, \Psi(w) \, \varphi(w) \, dw,$$

a limit in which no resolution appears. But the pair $(g_*\mu, \,\cdot\,)$ does not suffice downstairs: the function $a(x, \cdot)$, hence ξ, genuinely varies along positive-dimensional fibres of g, so the limit functional cannot in general be expressed on W. What is resolution-independent is the *law* of $\mathcal{Y}(\xi)$, since by Theorem 6.4 it is the limit in law of the resolution-free quantities $n^\lambda (\log n)^{1-m} Z_n^0$.

Remark 9.2 (What has been replaced). Definition 3.1 and Proposition 3.2 replace [1, Thm. 4.6] together with the consistency bookkeeping of [1, Thm. 6.5(3)–(4)]; Theorem 5.2 replaces [1, Thms. 4.7–4.9] and the chart-by-chart assembly in the proof of [1, Thm. 6.7], and covers all insertions at once. Convexity of the orthant charts makes the interpolation-to-the-face step elementary and avoids constructing a global C^1 retraction onto Σ (which would face gluing questions where essential faces meet). Charts persist inside the proofs of Proposition 3.2, Lemmas 4.1–4.4 and Theorem 5.2; the monomial evaluations there, formerly cited from [1, Thm. 4.6, Cor. 4.3], are now proved in place — the folding bound in the proof of Lemma 4.1, the logarithmic-coordinate computation of Lemma 4.4, and the Chebyshev display in Step 3 of the proof of Theorem 5.2 — while the chart-locality itself appears irreducible, the evaluation being a monomial computation. Two smaller repairs are worth recording. First, the error bound of [1, Thm. 4.9] involves $\|\nabla\xi\|$; in [1] the corresponding moment bound $\mathbb{E}\|\nabla\xi_n\|^s \le \text{const}$ is stated as [1, Lem. 6.4(1)], under $s = 6$ and with a one-line proof from [1, Thm. 5.8]. Lemma 6.1 spells out the argument — the gradient is itself the centred empirical process of an $L^2(q)$-valued analytic family, so [1, Thm. 5.8] applies to it — and provides the bound at index $s = 2$, where Sections 3–6 need it. Second, under the dictionary of Remark 3.3, $\mathcal{Y}(\xi)$ coincides with the chartwise limit $Y^{(1)}(\xi)$ appearing in the proof of [1, Thm. 6.7]; the limit displayed in the *statement* of [1, Thm. 6.7] omits the monomial factor y^μ that its own proof (and [1, Thm. 4.9]) carries.

Remark 9.3 (What remains). With Sections 7–8 in place, no estimate in the derivation of the four errors is imported from [1]: the last citation — the σ-tilted moment bounds and cubic Taylor remainders of [1, Lem. 6.5], displays (6.57), (6.59) in the proof of [1, Thm. 6.8] — is discharged by Theorem 8.18, Corollary 8.19 and Lemma 8.21. For clarity we list what *is* still taken from [1], all of it declared in §2 or Remark 9.2: the resolution package and the standard form ([1, Thms. 6.1, 6.5]); the empirical-process moment theory ([1, Thm. 5.8], used for ξ_n, ψ_n, $\nabla\xi_n$, and the auxiliary processes χ_n, $\tilde{\chi}_n$ of Lemma 8.23); the meromorphic continuation of the zeta function ([1, Thm. 6.6]); and generic probabilistic background (convergence of laws, asymptotic uniform integrability). (The chart-local monomial evaluations, formerly on this list, are now proved in §4.) The localization of the Bayes losses between the full and the inner posterior, for which [1] uses the H_t-tail machinery of [1, Lems. 6.1–6.3], is proved here in mean as Lemma 8.27: the sure envelope $|\log\langle e^{-f(x,\cdot)}\rangle| \le M(x)$ and the superexponential smallness of the outer posterior mass on $\{V_n \le n\epsilon/4\}$ together beat the factor $e^{2M(x)}$ for all $M(x) = O(n)$, and polynomial moments handle the complements. The Gaussian integration by parts formerly cited from [1, Thm. 5.11, eq. (5.24)] is now proved as Theorem 8.13, making the probabilistic layer of the limit theory self-contained. Everything else in the derivation of the four formulas is proved here. Finally, maximum likelihood and maximum a posteriori estimation lie outside not merely the Bayesian scope of this note but the method itself; see Remark 9.4.

Remark 9.4 (The boundary of the method). A fair one-sentence summary of Sections 3–8 is that everything came free where the sample enters the Gibbs weight only through the canonical field $\beta\sqrt{t}\,\xi_n(u)$, with all further sample dependence confined to C^1 prefactors — and that everything that resists is a place where the sample enters the weight beyond it. More precisely, the toolkit consists of three mechanisms. (i) *Insertions against the canonical weight*: any $F \in \mathcal{P}_A$ — polynomially bounded in t, C^1 in u, depending on the sample only through prefactors such as ξ_n, $a(x,\cdot)$, or the auxiliary empirical processes of Lemma 8.23 — is handled by Theorem 5.2 and Proposition 8.4, in one or two replicas; this yields the statements in law. (ii) *Free energies and their first coupling-derivatives*: since (λ, m) are independent of the couplings and of β, the $\lambda \log n$ terms cancel in differences, and convexity converts the envelope bounds of Lemma 7.4 into sure polynomial envelopes for first derivatives (Theorem 8.15, Lemma 8.17); together with the sure loss envelope $|\log\langle e^{-f(x,\cdot)}\rangle| \le M(x)$ of Lemma 8.27 and the uniform integrability background of Theorem 7.2, this yields the statements in mean. (iii) *The Gaussian calculus of the limit functionals* on $(\Sigma, \mu, \xi|_\Sigma)$ (Theorem 8.13, Remark 8.14), self-contained and independent of the kernel theorem.

The dividing line is a matter of envelopes. Statements in law tolerate the factors $e^{2\beta\|\xi_n\|_\infty^2}$ of Proposition 8.4; statements in mean require polynomial envelopes, and the toolkit produces these only for logarithms of partition functions and for their first coupling-derivatives. For the second derivative, convexity yields merely an averaged statement: there is some $\gamma^* \in (1, \frac{3}{2})$ with $\mathrm{Var}_{\gamma^*}\big(\beta(t - \sqrt{t}\,\xi_n)\big) \le C\,(1 + \|\xi_n\|_\infty^2)$, surely for $n \ge n_0$, at an uncontrolled nearby coupling rather than at $\gamma = 1$. Every resistant item of [1, §6.3] reduces to pointwise polynomial envelopes for the un-logged moments $\langle t^k\rangle_n$, $k \ge 2$ — and these, it turns out, the toolkit does supply. Theorem 8.18 proves them, not by a finite-n integration by parts (the vector field implementing ∂_t upstairs is the normalized chartwise Euler field $\frac{1}{m}\sum_{j\le m}(x_j/2k_j)\,\partial_{x_j}$, exactly the kind of chart-dependent object whose gluing the present formulation is designed to avoid, and no surrogate for it is constructed) but by combining the nonnegativity of t with exponential insertions of optimized vanishing rate, whose logarithms are short coupling-differences of free energies — mechanism (ii) again. The averaged-variance limitation is thereby bypassed rather than overcome: no pointwise second-derivative bound is ever established, and none is needed. Consequently the σ-tilts of [1, Lem. 6.5], which are $O(n^{-1/2}\sqrt{t}\,M(x))$ perturbations of the canonical weight, are controlled uniformly by Corollary 8.19; the Bayes errors' interpolation $\sigma \mapsto -\sum_i \log\langle e^{-\sigma f(X_i,\cdot)}\rangle$ has interior derivatives that are exactly such tilted averages, while its endpoint slopes give only Jensen's inequality $B_t \le G_t$. The third-order Taylor bookkeeping is carried out in Lemma 8.21, completing the removal: the four-error derivation imports no estimate beyond the declared background. The boundary is thus internally consistent: statements chart-free, proofs chart-local — and Theorem 8.18 is precisely the statement-level reformulation of the content of [1, Lem. 6.5] whose absence this remark previously recorded, obtained, as it happens, without any new chart computation.

Finally, maximum likelihood and maximum a posteriori estimation lie outside the method altogether, not merely outside its Bayesian scope: their errors are governed by suprema of the empirical process over the resolved space, nothing is integrated against ω, and neither μ nor (λ, m) enters — sup-functionals do not see the leading measure.

Remark 9.5 (Literature). The scheme "coefficients of the asymptotic expansion of $\int e^{-nf}\varphi$ are distributions supported on the critical locus, realized as Laurent coefficients of $\int f^z\varphi$, with orders controlled by the drop in pole order" is developed in Arnold–Gusein-Zade–Varchenko [3, Vol. II, Part II], including Varchenko's computation of (λ, m) from Newton polyhedra in nondegenerate cases [4]; two caveats when consulting these sources are that the amplitudes there are usually nonvanishing (the monomial density $u^h\phi$ shifts the numerology exactly as in (8)), and that much of the theory is stated for isolated critical points. Malgrange [5] connects the pole structure to the Bernstein–Sato polynomial, i.e. to [1, §4.5]; Barlet [6] is the complex-analytic counterpart, where the positivity of leading coefficients appears in the theory of the currents associated with $\int |f|^{2z}$.

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